

Bronchoscopic Lung Volume Reduction with Endobronchial Valves: A Consensus Statement on Practical Aspects of Patient Selection and Periprocedural Management

Momen M. Wahidi^a Robert J. Lentz^{b,c} Gerard J. Criner^d
Daniela Gompelmann^e Alvin J. Ing^f Christopher M. Kapp^a Shiyue Li^g
Frank C. Scurba^h Pallav L. Shah^{i,j} Dirk-Jan Slebos^{k,l} Charlie Strange^m
Felix J. Herth^{n,o}

^aDivision of Pulmonary and Critical Care, Northwestern Feinberg School of Medicine, Chicago, IL, USA;

^bDivision of Allergy, Pulmonary, and Critical Care Medicine, Vanderbilt University Medical Center, Nashville, TN, USA; ^cDepartment of Thoracic Surgery, Vanderbilt University Medical Center, Nashville, TN, USA; ^dDepartment of Thoracic Medicine and Surgery, Lewis Katz School of Medicine at Temple University, Philadelphia, PA, USA;

^eDivision of Pulmonology, Department of Internal Medicine II, Medical University of Vienna, Vienna, Austria;

^fFaculty of Medicine, Health and Human Sciences, Macquarie University, Sydney, NSW, Australia; ^gGuangzhou Institute of Respiratory Health, State Key Laboratory of Respiratory Disease, National Clinical Research Center for Respiratory Disease, National Center for Respiratory Medicine, The First Affiliated Hospital of Guangzhou Medical University, Guangzhou, China; ^hDivision of Pulmonary, Allergy and Critical Care Medicine, University of Pittsburgh, Pittsburgh, PA, USA; ⁱRoyal Brompton Hospital, Department of Pulmonology, London, UK; ^jNational Heart and Lung Institute, Imperial College, London, UK; ^kDepartment of Pulmonary Diseases, University of Groningen, University Medical Centre Groningen, Groningen, The Netherlands; ^lGroningen Research Institute for Asthma and COPD, University of Groningen, University Medical Centre Groningen, Groningen, The Netherlands;

^mDivision of Pulmonary, Critical Care, Allergy, and Sleep Medicine, Medical University of South Carolina, Charleston, SC, USA; ⁿDepartment of Pneumology and Critical Care Medicine, Thoraxklinik at the University Hospital Heidelberg, Heidelberg, Germany; ^oTranslational Lung Research Center Heidelberg (TLRC) and German Center for Lung Research (DZL), Heidelberg, Germany

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Abstract

Endobronchial valve implantation to accomplish bronchoscopic lung volume reduction (BLVR) has been shown to improve pulmonary function, functional status, and dysp-

nea in patients with advanced emphysema complicated by hyperinflation without interlobar collateral ventilation.

Momen M. Wahidi and Robert J. Lentz contributed equally to this work.

Trial registration is not available for the work.

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Patient selection and periprocedural management are largely derived from clinical trial inclusion criteria and limited to expert opinion. No consensus recommendations are available to guide practical aspects of BLVR. An international group of experts were invited to participate and were surveyed broadly regarding their BLVR-related practices, from which specific topics were selected by the group, followed by literature review on these topics. Recommendations regarding selected topics were formulated over three voting rounds using a modified Delphi technique requiring at least 80% consensus agreement. A total of 21 recommendations reached consensus and are detailed herein. One recommendation failed to reach consensus, and two other topics were found to have insufficient evidence to make specific recommendations. These consensus recommendations offer guidance on practical aspects of patient selection and periprocedural management topics related to BLVR with endobronchial valves and highlight gaps in current knowledge requiring further research.

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Summary Statements

Pre-Procedure: Patient and Target Lobe Selection

1. In patients with advanced, hyperinflated emphysema being considered for bronchoscopic lung volume reduction (BLVR), we suggest multidisciplinary review of clinical, radiological, and physiological data when local expertise in chronic obstructive pulmonary disease, lung volume reduction surgery, and/or lung transplant is available.
2. In patients with advanced, hyperinflated emphysema being considered for BLVR, we suggest proceeding with caution when ≥ 4 liters per minute supplemental oxygen is required to maintain adequate oxygen saturation at rest.
3. In patients with advanced, hyperinflated emphysema being considered for BLVR, we recommend against proceeding when arterial partial pressure of carbon dioxide exceeds 55 mm Hg despite optimal preoperative medical therapy and nocturnal ventilation.
4. In patients with advanced, hyperinflated emphysema being considered for BLVR, we suggest proceeding with caution in the presence of systolic dysfunction (left ventricular ejection fraction $< 40\%$). We recommend against proceeding if there is clinical evidence of decompensated heart failure.
5. In patients with advanced, hyperinflated emphysema being considered for BLVR, we suggest proceeding with caution in the presence of moderate pulmonary hypertension and recommend against proceeding if pulmonary hypertension is severe.
6. In patients with advanced, hyperinflated emphysema being considered for BLVR, we recommend against exclusion of patients based on high or low body mass index (BMI). Patients with habitus-associated comorbid conditions (e.g., obesity-hypoventilation, malnutrition or pulmonary cachexia) or very high or low BMIs (e.g., < 18 , > 40) should be approached with caution.
7. In patients with advanced, hyperinflated emphysema being considered for BLVR, we suggest evaluation of patients with either heterogeneous or homogeneous distribution of centrilobular or panlobular emphysema. We suggest avoiding routine BLVR in those with predominantly paraseptal emphysema or giant bulla.
8. In patients with advanced, hyperinflated emphysema being considered for BLVR, we suggest avoiding valve placement ipsilateral to prior lobectomy or prior pleurodesis. Valve placement may be considered in the setting of prior contralateral lobectomy or pleurodesis, prior ipsilateral sub-lobar resection, or after sternotomy.
9. In patients with advanced, hyperinflated emphysema being considered for BLVR, we recommend against excluding patients based on alpha-1 antitrypsin deficiency.
10. In patients with advanced, hyperinflated emphysema being considered for BLVR, we suggest using quantitative computed tomography (CT) analysis to estimate fissure integrity, extent of lobar emphysematous destruction, and lobar volumes in all patients to inform procedure candidacy and lobe targeting. We suggest lobes with extensive emphysema ($\geq 50\%$ low attenuation score at -910 Hounsfield units (HU) or $\geq 20\%$ at -950 HU) adjacent to fissures that are at least 80% intact be considered potential targets for valve placement.
11. In patients with advanced, hyperinflated emphysema being considered for BLVR who have multiple potential lobar targets or a homogeneous disease distribution, we suggest consideration of pre-procedure perfusion testing.
12. In patients with advanced, hyperinflated emphysema being considered for BLVR, targeting lobe(s) in a lung with significantly lower overall emphysema scores or greater perfusion should be avoided.

13. In patients with advanced, hyperinflated emphysema being considered for BLVR who have multiple potential lobar targets (see statement #10), we suggest using emphysema destruction scores, lobe volume, lobar perfusion, and presence of disqualifying incidental findings to determine the primary target lobe.
14. In patients with advanced, hyperinflated emphysema being considered for BLVR, we recommend against excluding patients based on a perceived increased risk of post-procedure pneumothorax.
21. In patients with advanced, hyperinflated emphysema who experience a marked increase in supplemental oxygen requirement in the perioperative period, we suggest evaluation for reversible etiologies, such as pneumothorax, pneumonia, or pulmonary emboli.

Procedural Topics

15. In patients with advanced, hyperinflated emphysema undergoing bronchoscopy for BLVR, we suggest performing a direct physiological assessment for collateral ventilation (e.g., bronchoscopic balloon occlusion) before placing valves in the left lung if fissure integrity score is <95% and prior to all right-sided valve placements.
16. In patients with advanced, hyperinflated emphysema who undergo BLVR, we recommend post-procedure hospital admission for a minimum of three nights.

Post-Procedure Management Topics

17. In patients with advanced, hyperinflated emphysema who have undergone BLVR, we suggest patients be evaluated for clinical/functional status, radiographic response, and complications at 4–6 weeks, 3–6 months, and 12 months after procedure.
18. In patients with advanced, hyperinflated emphysema with subtotal atelectasis of the target lobe without clinical improvement by 3 months post-BLVR, we suggest evaluation with CT chest and consideration of bronchoscopic inspection with valve revision or removal if indicated.
19. In patients with advanced, hyperinflated emphysema who develop complete atelectasis of the target lobe without clinical improvement by 3 months post-BLVR, we suggest PFTs, CT chest, cardiac evaluation, and other testing guided by clinical evaluation.
20. In patients with advanced, hyperinflated emphysema who experience a clinical and radiographic response post-BLVR followed by loss of clinical and radiographic response, we suggest CT chest and revision bronchoscopy to evaluate for valve malposition or malfunction, granulation tissue, infection, or other etiology.

Introduction

Chronic obstructive pulmonary disease (COPD) is a significant cause of morbidity and mortality globally, with disease burden expected to increase in coming decades [1, 2]. COPD is a heterogeneous disease process. One major endotype of COPD, emphysema, results from destruction and enlargement of airspaces distal to the terminal bronchioles, resulting in loss of elasticity [3]. Subsequent air trapping leads to hyperinflation, which is associated with worse performance status, increased breathlessness, impaired cardiac function, more acute disease exacerbations, and increased mortality [4–8].

Management of emphysema ranges from noninvasive modalities like pharmacotherapy, pulmonary rehabilitation, and supplemental oxygen to procedural interventions including bronchoscopic lung volume reduction (BLVR), lung volume reduction surgery (LVRS), and lung transplantation [9–12]. BLVR has been shown to improve pulmonary function, functional status, and dyspnea in large randomized trials and is now included in the Global Initiative for Chronic Lung Disease (GOLD) management guidelines. BLVR utilization remains low, performed in only an estimated 5% of patients with emphysema complicated by severe air trapping, but is expected to rise [13–16].

As BLVR utilization increases, there is a need to generate consensus management recommendations informed by available published data and current practice at high volume centers. This management consensus statement will provide context on the current literature surrounding patient selection, periprocedural considerations, and post-operative management.

Methods

Eleven BLVR experts were invited and all agreed to participate. Experts were selected based on extensive experience (e.g., >100 procedures, Principal Investigators in large BLVR randomized trials, extensive COPD expertise) with emphasis on geographic and practice-setting diversity.

All experts completed an initial survey (see online suppl. material; for all online suppl. material, see <https://doi.org/10.1159/000548437>) to indicate the BLVR-related topics they felt most pertinent to practical patient selection and management, regardless of the perceived level of published evidence available. Responses were reviewed during a virtual meeting attended by all participants and topics were selected. Initial recommendation statements were drafted by R.J.L. and M.M.W. in patient, intervention, comparison, and outcome (PICO) format. Each PICO statement was assigned to two experts tasked with performing independent literature searches on each topic, followed by modification of the draft recommendation as necessary; these modified recommendation statements served as the starting point for Delphi voting rounds.

A modified Delphi methodology was agreed upon during the virtual meeting. Delphi surveys were conducted in REDCap [17, 18] with unique URLs sent to each participant to prevent duplicate responses and ensure 100% participation in all rounds while maintaining anonymity. Respondents indicated their level of agreement on all recommendations using a five-point Likert scale ranging from “strongly disagree” to “strongly agree.” Respondents not in agreement were required to describe changes necessary to gain their agreement. Consensus was defined as >80% of respondents answering “agree” or “strongly agree” to a given recommendation. After each round, recommendations not meeting consensus were modified by R.J.L. and M.M.W. based on comments and included in the next round. Inability to obtain consensus was declared if a recommendation failed to reach consensus after three Delphi voting rounds.

Results

Three Delphi voting rounds were completed between January 16, 2024, and June 4, 2024. All eleven voting experts (all authors except C.M.K.) participated in each round with 100% complete surveys. A total of 23 recommendations were included in the first voting round. Thirteen recommendations reached consensus in the first round, three in the second round, and five in the third round (Fig. 1, 2). Consensus could not be reached on one topic (pulmonary rehabilitation) and there was agreement that available evidence was insufficient to support a recommendation on two additional topics (BLVR staged over two procedures, sequential bilateral

BLVR). One original recommendation was split into two distinct recommendations which both reached consensus in the third round (#4, 5). Full text of recommendation statements in each round is included in the online supplementary material.

Pre-Procedure: Patient and Target Lobe Selection

The following recommendations discuss patient factors, radiological findings, and physiological testing, making patients more or less optimal BLVR candidates. Randomized controlled trial inclusion and exclusion criteria (summarized in online suppl. Table S1), which are often conservative to facilitate research goals, underlie most current BLVR candidacy decisions. The threshold at which any specific derangement should be considered an absolute contraindication is controversial or undefined in the literature for most elements. Instead, the panel encourages a nuanced, integrated assessment that considers all the following aspects when assessing BLVR candidacy. Patients who are borderline in multiple domains may ultimately be deemed poor candidates, while patients with a single significantly aberrant physiological element may still be deemed candidates, with appropriate discussion of risks and benefits according to the situation, particularly at more experienced treating centers. Therefore, the following recommendations should be considered clinical guidance but not a list of relative or absolute contraindications for this procedure. Shared decision-making with patients is encouraged.

The main rationale for caution in the setting of significantly deranged physiological testing is concern for tolerance of sudden-onset complications like pneumothorax, which occurs commonly in the periprocedural period. Additionally, some patient factors discussed may reduce the magnitude of benefit.

Recommendation 1

- In patients with advanced, hyperinflated emphysema being considered for BLVR, we suggest multidisciplinary review of clinical, radiological, and physiological data when local expertise in COPD, LVRS, and/or lung transplant are available.

Literature Review

COPD is a complex disease with expanding medical and interventional therapeutic options. In patients with hyperinflation, LVRS and lung transplant are considerations in addition to BLVR. In other complex diseases, including interstitial lung disease and lung

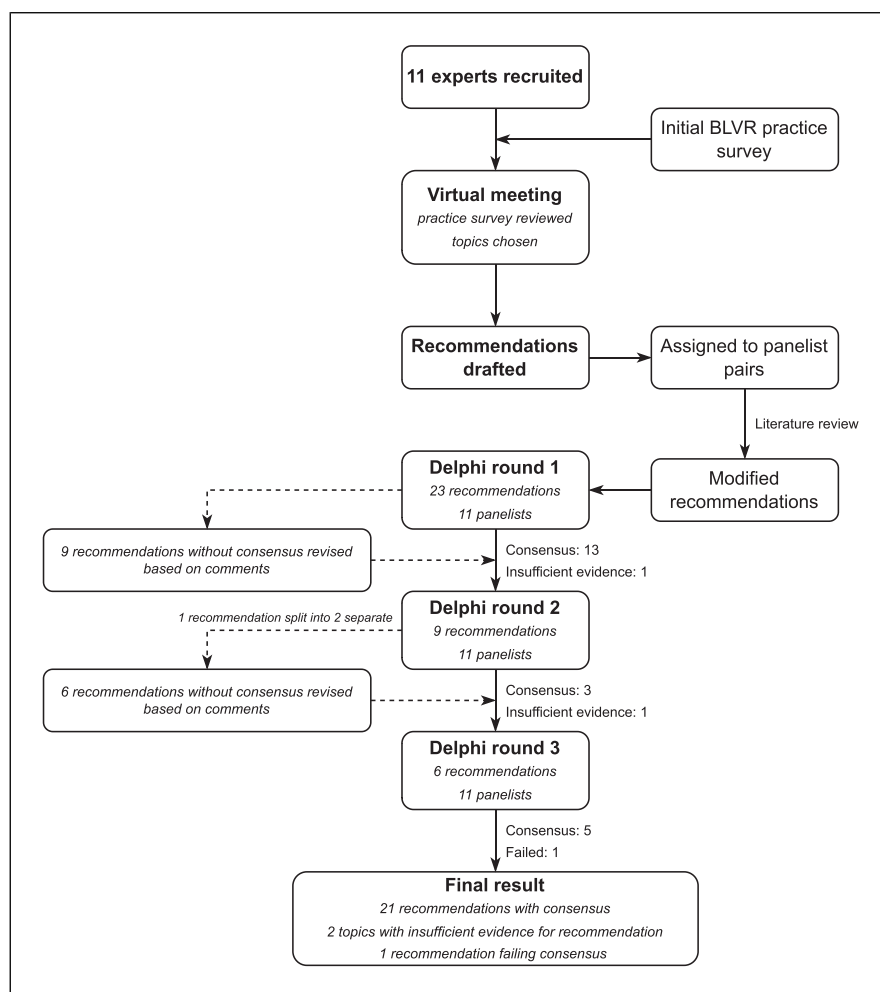


Fig. 1. Consensus process flow.

cancer, multidisciplinary committees (MDC) have been shown to reduce heterogeneity in treatment recommendations and improve outcomes [19, 20]. The UK National Institute for Health and Care Excellence and multiple expert reviews have recommended MDC in patient selection for lung volume reduction and several groups have published their experience in establishing an MDC review, though clinical outcomes data are lacking [21–26]. One major BLVR randomized controlled trial, BeLieVeR-HiFi, utilized MDC in patient selection decisions, but the study did not specifically address its impact [27].

Rationale for Recommendations

MDC review of potential candidates for BLVR offers several benefits: many patients eligible for BLVR are also eligible for LVRS and/or lung transplant, and case review by thoracic surgeons and transplant pulmonologists may identify patients

better suited for these interventions; pulmonologists with expertise in COPD can optimally assess adequacy of medical therapy; multidisciplinary radiological review may identify otherwise overlooked lung nodules, bronchiectasis, or diaphragmatic pathology, among other potential benefits. MDC participants may include interventional pulmonologists or advanced bronchoscopists, pulmonologists with expertise in COPD, transplant pulmonologists, thoracic surgeons with expertise in LVRS and/or lung transplant, thoracic radiologists, and pulmonary rehabilitation specialists, depending on availability of local expertise.

However, outcomes data are limited to date. In the absence of a clear demonstration of improved outcomes with MDC compared to without, MDC should not be universally required prior to offering BLVR to a candidate deemed eligible by a thoughtful unilateral review.

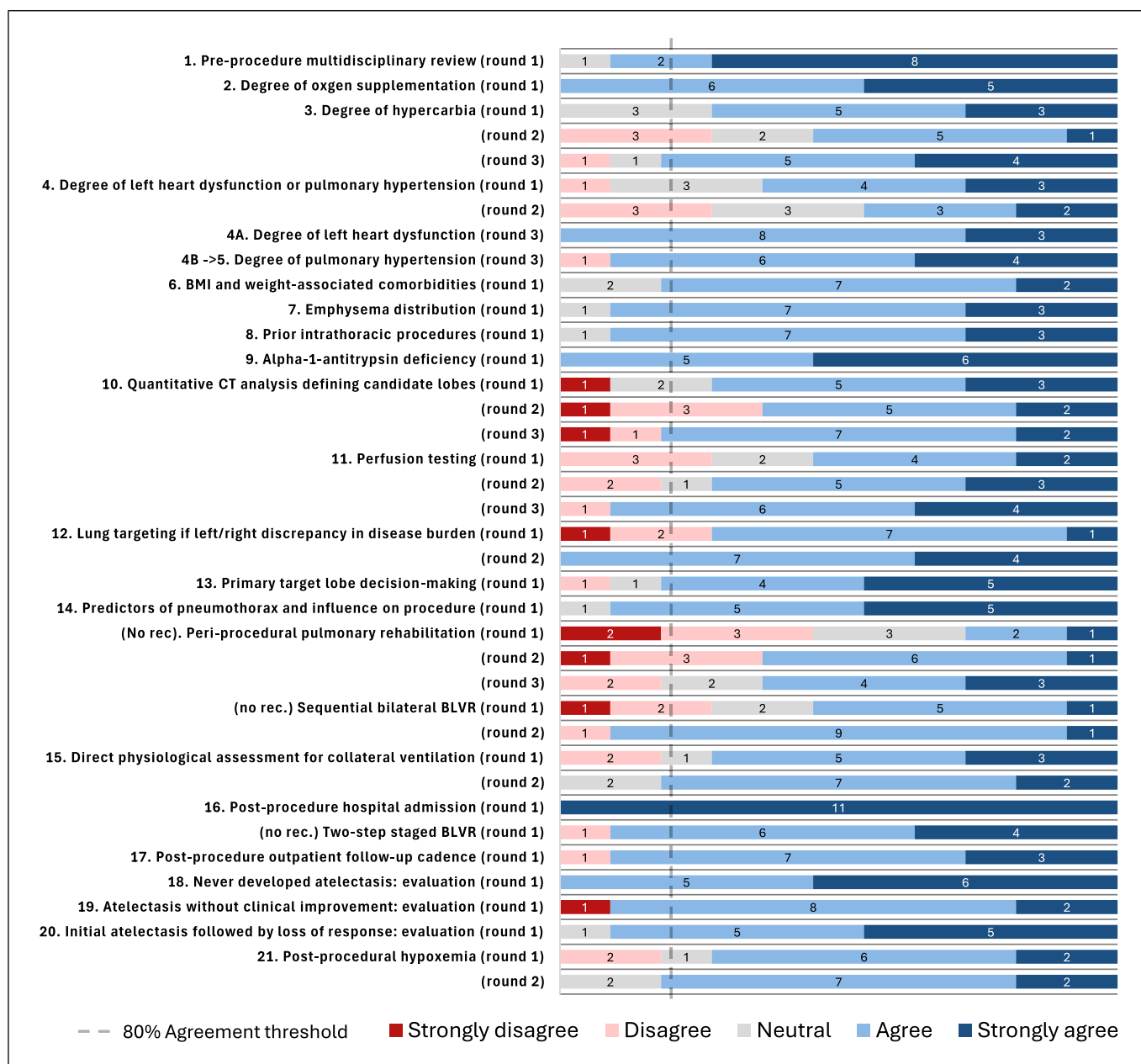


Fig. 2. Delphi voting round results. "No rec." indicates "no recommendation" because the statement did not achieve consensus through three rounds or there was consensus on insufficient data to support a recommendation. See online supplementary data for full text of each recommendation statement in each round.

Recommendation 2

- In patients with advanced, hyperinflated emphysema being considered for BLVR, we suggest proceeding with caution when ≥ 4 liters per minute supplemental oxygen is required to maintain adequate oxygen saturation at rest.

Literature Review

Hypoxemia is common in COPD, resulting from V/Q mismatch and loss of surface area for gas exchange [28]. Supplemental oxygen for patients with resting hypoxemia is one of few treatments in advanced COPD to demonstrate a mortality benefit [29, 30]. Major BLVR trials all excluded prospective candidates with severe

hypoxemia, defined in most as resting room air arterial partial pressure of oxygen <45 mm Hg [14, 15, 27, 31, 32]. A threshold of hypoxemia severity beyond which clinical outcomes worsen has not been well defined. No trials utilized supplemental oxygen flow rate as an eligibility criterion.

Rationale for Recommendations

A threshold of hypoxemia severity likely exists beyond which risk increases and clinical outcomes worsen, particularly should pneumothorax occur, as it commonly does in the BLVR perioperative period. That threshold has not been clearly delineated. Most major trials used a resting room air arterial partial pressure of oxygen threshold of 45 mm Hg, which corresponds to an oxygen saturation of ~80% at sea level under normal physiological conditions [28].

While supplemental oxygen flow rates have not been used to determine eligibility in major BLVR trials, they are more readily available in clinical practice and may be particularly helpful when screening patients for potential candidacy. Supplemental oxygen requirements should be titrated in a clinical setting to ensure they accurately reflect flow rates required to maintain an oxygen saturation of >88%.

Emphysema patients requiring high flow rates of supplemental oxygen often have additional comorbid conditions, including combined pulmonary fibrosis and emphysema, pulmonary hypertension, severe sleep apnea, or significant chronic bronchitis. These comorbidities should be evaluated to determine if potentially reversible contributions exist prior to considering BLVR.

Recommendation 3

- In patients with advanced, hyperinflated emphysema being considered for BLVR, we recommend against proceeding when arterial partial pressure of carbon dioxide (PaCO₂) exceeds 55 mm Hg despite optimal preoperative medical therapy and nocturnal ventilation.

Remarks: a well-established PaCO₂ threshold beyond which risk increases has not been established. Randomized clinical trials to date have excluded those with significant chronic hypercapnic respiratory failure, most using thresholds of 50–55 mm Hg.

Literature Review

Most major BLVR trials exclude those with PaCO₂ >50 or >55 mm Hg. A single trial (STELVIO) used a higher threshold of 8.0 kPa (60 mm Hg);

BeLieVeR-HiFi did not utilize a hypercarbia exclusion and REACH excluded patients with “severe gas exchange abnormality” not further specified; it is not clear how many patients in these trials had PaCO₂ >55 (online suppl. Table S1) [27, 32, 33]. Lenga and colleagues, in an analysis of prospectively collected registry data, found similar BLVR outcomes in those with baseline PaCO₂ ≤45 compared to those >45 mm Hg; mean PaCO₂ was 52 mm Hg in the higher group and decreased an average of 8 mm Hg by 3 months post-BLVR [34]. Another retrospective, single center analysis of 129 consecutive patients with baseline PaCO₂ >45 mm Hg undergoing BLVR did not report any deaths or episodes of severe respiratory failure; pneumothorax occurred in 16% and all recovered. Baseline mean PaCO₂ was 49.7 with improvement by an average of 3.7 points 3 months post-procedure [35].

Rationale for Recommendations

Most patients who have undergone BLVR reported in the literature had pre-procedure PaCO₂ <55 mm Hg. Publications focusing on patients with greater degrees of hypercarbia have reported mean PaCO₂ below this threshold. Similar to advanced hypoxemia, a hypercarbia threshold beyond which procedural risk increases and/or benefit wanes has not been well defined. A threshold of 55 mm Hg is, therefore, prudent pending clarifying evidence.

Nocturnal noninvasive ventilation is a consideration in any COPD patient with chronic hypercapnic respiratory failure and may be pursued in patients with PaCO₂ >55 mm Hg to determine if this optimization will lower their PaCO₂ into a range more acceptable for BLVR consideration. When severely elevated PaCO₂ is the only relative contraindication in an otherwise ideal candidate, BLVR may be considered but should be accompanied by a detailed discussion of risks and benefits.

Recommendation 4

- In patients with advanced, hyperinflated emphysema being considered for BLVR, we suggest proceeding with caution in the presence of significant systolic dysfunction (left ventricular ejection fraction [LVEF] <40%). We recommend against proceeding if there is clinical evidence of decompensated heart failure.

Remarks: patients with well-compensated heart failure on stable medical regimens and LVEF ≥40% should continue their BLVR candidacy evaluation. BLVR should be avoided in those with decompensated heart failure, whether reduced or preserved ejection fraction.

Literature Review

COPD is associated with a higher incidence of cardiac disease than in the general population. In a recent analysis of National Health and Nutrition Examination Survey data ($n = 31,560$), 15–19% of individuals with emphysema had comorbid heart failure, a prevalence 7 times that of non-COPD patients [36]. COPD patients with comorbid heart failure, including heart failure with preserved ejection fraction, are at increased risk of hospitalization and death [37, 38]. Systolic dysfunction is associated with an increased risk of perioperative mortality, myocardial infarction, and heart failure exacerbation [39].

VENT and LIBERATE trials excluded patients with LVEF <45%; most others included a generic criterion excluding patients with any major comorbidity which limited exercise capacity, which would likely have excluded patients with clinical heart failure. Limited data have been published regarding the safety or efficacy of BLVR in patients with lower LVEFs or clinical heart failure.

The effect of BLVR on cardiac function has been investigated in two small prospective observational studies involving patients without known cardiac disease undergoing BLVR. Van der Molen and colleagues [40] evaluated 24 patients with cardiac MRI 1 day before and 8 weeks after BLVR, noting significant improvements in stroke volume of both ventricles, preload, and cardiac output. Pizarro and colleagues evaluated patients with transthoracic echocardiography 1 day before and 8 weeks after BLVR; 16 clinical responders were matched to 16 non-responders. Left-sided indices did not change after successful BLVR, but right ventricular strain parameters improved in responders [41].

Rationale for Recommendations

There is limited published literature regarding BLVR safety or efficacy in patients with significant comorbid heart failure as these patients were effectively excluded from all major BLVR trials. An LVEF threshold beyond which risk (predominantly perioperative) increases or benefit is reduced has not been described. In this context, it seems prudent to rely on clinical signs and symptoms; patients with a history of heart failure (with reduced or preserved LVEF) with minimal heart failure-specific signs or symptoms on stable medical regimens should continue their candidacy evaluations. A careful cardiac history and examination should be undertaken in those with LVEF <40%, including appropriate additional cardiac evaluations if not already performed. Cardio-pulmonary exercise testing is a consideration in those

with reduced LVEF to clarify cardiac contributions to the patient's functional status or confirm a ventilatory limitation.

While there is some evidence that successful BLVR may improve various cardiac indices in small observational studies, the clinical significance of these changes and whether they also occur in those with underlying heart failure has not yet been established.

Recommendation 5

- In patients with advanced, hyperinflated emphysema being considered for BLVR, we suggest proceeding with caution in the presence of moderate pulmonary hypertension and recommend against proceeding if pulmonary hypertension is severe.

Literature Review

Pulmonary hypertension is a common complication of advanced COPD, with a prevalence of up to 40%, and is associated with worse survival [42–44]. In COPD patients, most pulmonary hypertension is mild or moderate (mPAP 20–35), with severe disease (mPAP >35) present in 7% in one recent meta-analysis [44]. Various severity grading schemas have been proposed to define severe COPD-associated pulmonary hypertension, including pulmonary vascular resistance >5 Wood units (used in recent ERS/ESC guidelines), mean PA pressure ≥ 35 mm Hg, or associated with right ventricular systolic dysfunction [45, 46].

There are limited data specific to BLVR in patients with known pre-existing pulmonary hypertension. Most major BLVR RCTs excluded patients with significant pulmonary hypertension, either via specific pressure cut-offs (e.g., PASP >45 in LIBERATE and IMPACT) or due to “known” or “severe” pulmonary hypertension or presence of cor pulmonale [14, 15, 31, 47, 48]. In a small prospective observational series of 6 patients with pulmonary hypertension who underwent RHC before and after BLVR, Eberhardt and colleagues reported no major complications [49]. Mean PA pressure before BLVR was 31 ± 6.5 mm Hg (maximum, 43). Five of six had lower mPAP 90 days post-BLVR (mean 28.5 ± 5.1) while one had higher mPAP (28–31 mm Hg). PAWP and PVR also reduced slightly, and cardiac index improved slightly. In a retrospective single center series, among 375 treatment patients, 13 were noted to have mildly elevated estimated mean pulmonary artery pressures (defined as ≥ 25 mm Hg; maximum was 30) on pre-procedure transthoracic echocardiogram [50]. There were no observed cardiac complications. Among those achieving significant

volume loss, estimated mean pulmonary artery pressure decreased from mean 25.9 to 22.3 mm Hg, of uncertain clinical significance.

In the LVRS literature, there are reports of pulmonary hypertension developing after LVRS, hypothesized to be related to a reduction in pulmonary vascular capacitance due to resection of part of the pulmonary vascular bed [51]. Criner and colleagues [52] in a sub-study of the National Emphysema Treatment Trial (NETT) including 55 patients with RHC data before and after LVRS, reported no mean increase in PA pressures at rest after LVRS. Thuppal and colleagues [53] in a single center retrospective study, compared LVRS outcomes in 48 patients with mild or moderate pulmonary hypertension and 8 with severe pulmonary hypertension to 68 without pulmonary hypertension and found similar clinical outcomes in both groups. Haniuda and colleagues [54] compared RHC data on patients before and after LVRS ($n = 7$) and lobectomy ($n = 8$), respectively, noting an increase in pulmonary vascular resistance after both surgeries, particularly during exercise, but increased PA pressures observed after lobectomy did not occur after LVRS and exercise-induced increase in PAWP was lower in magnitude after LVRS compared to lobectomy. Another group, in a small observational series of LVRS patients with baseline PA systolic pressure >35 , noted a mean decrease in this pressure after LVRS (41–37 mm Hg), with no complications attributed to pulmonary hypertension [55].

Rationale for Recommendations

There are limited published data regarding the safety and efficacy of BLVR in patients with significant pulmonary hypertension. Most pertinent data are derived from hemodynamic studies of LVRS patients. While these generally did not find significant mean increases in right-sided pressures after lung volume reduction and some reports suggest pressures and RV function may improve post-lung volume reduction, these studies had modest numbers of participants. Additionally, while LVRS is similar in concept, it is not a precisely analogous procedure.

The panel suggests assessing for pulmonary hypertension in all potential BLVR candidates. Hypoxemia out of proportion to emphysema burden, syncope or near syncope, peripheral edema, elevated jugular venous pressures, loud or split pulmonic second heart sound, elevated serum BNP, or TTE findings of elevated right-sided pressures, RV dilation, or RV systolic dysfunction should all prompt pulmonary hypertension

evaluation. In lieu of well-defined right-sided pressure thresholds associated with higher risk or reduced benefit, the panel suggests caution when significant pulmonary hypertension or pulmonary hypertension associated with right ventricular dysfunction is present. Right heart catheterization should be considered prior to ruling a patient out as a candidate based on TTE findings; in a recent series of patients being evaluated for BLVR, most with TTE-estimated RVSP >45 mm Hg did not have severe pulmonary hypertension at RHC [56].

Recommendation 6

- In patients with advanced, hyperinflated emphysema being considered for BLVR, we recommend against exclusion of patients based on high or low BMI. Patients with habitus-associated comorbid conditions (e.g., obesity-hypoventilation, malnutrition or pulmonary cachexia) or very high or low BMIs (e.g., <18 , >40) should be approached with caution.

Literature Review

There are no specific data associating BMI, whether high or low, with BLVR-related outcomes. Two major RCTs featured exclusions for high BMI (VENT: >31.1 (men) or >32.3 (women); LIBERATE: >35) and two excluded for low BMI or significant weight loss (EMPROVE: BMI <15 ; LIBERATE: unplanned loss of $>10\%$ body weight in prior 3 months) [14, 15, 31]. Mean BMI in all major trials ranged 22–25.

Pulmonary cachexia is a well-recognized manifestation of late-stage COPD. BMI lower than 21 worsens prognosis in the widely used BODE index of COPD survival. Sarcopenia, associated with low BMI with a prevalence of 22% in patients with COPD (versus 10% among all older adults), is associated with worse prognosis [57, 58]. Morbid obesity is associated with pulmonary restriction, sleep apnea, and obesity-hypoventilation syndrome, affecting physiological respiratory reserve and complicating anesthetic management. High BMI also decreases the likelihood of having a residual volume by plethysmography sufficiently high to consider BLVR.

Data regarding the effect of lung volume reduction on BMI are mostly derived from the LVRS literature, as summarized in a systematic review by ter Haar and colleagues [59]; among 12 studies of BMI change after LVRS including 799 patients, most reported increased BMI post-procedure. However, in two studies of BMI post-BLVR involving 33 patients, there was no significant change in BMI post-procedure. Two studies, one

BLVR and one LVRS, which stratified by baseline BMI, found significant gains in those with low baseline BMIs while those with higher baseline BMIs lost weight [60, 61].

Rationale for Recommendations

Most patients undergoing BLVR reported in major trials had normal BMIs, and several trials specifically excluded patients with high or low BMI. Very high BMIs might reduce the potential benefit of BLVR as excessive weight contributing to dyspnea would remain, and/or increase risk by affecting ventilation, complicating airway management, or making chest tube placement in the event of significant periprocedural pneumothorax more challenging. Very low BMI or ongoing abnormal weight loss is associated with sarcopenia, frailty, and worse COPD-related prognosis; while the LVRS literature suggests BMI tends to improve post-procedure in low BMI patients, this has not yet been described in the BLVR literature. In the absence of specific data suggesting thresholds for BMI-related safety, the panel suggests assessing primarily for weight-related complications reasonably expected to increase risk or reduce benefit (pulmonary cachexia, obesity-hypoventilation syndrome) and proceed with caution at the extremes of BMI.

Recommendation 7

- In patients with advanced, hyperinflated emphysema being considered for BLVR, we suggest evaluation of patients with either heterogeneous or homogeneous distribution of centrilobular or panlobular emphysema. We suggest avoiding routine BLVR in those with predominantly paraseptal emphysema or giant bulla.

Literature Review

Different aspects of the secondary pulmonary lobule may be disproportionately affected by the parenchymal destruction that defines pulmonary emphysema, resulting in distinct emphysema subtypes based on disease distribution [62, 63]. Centrilobular (centriacinar) emphysema is the most common, highly associated with cigarette smoking and other chronic noxious inhalational exposures, results from destruction focused at the level of the proximal respiratory bronchioles. Panlobular emphysema involves uniform destruction across the secondary pulmonary lobule and is often associated with alpha-1 antitrypsin (AAT) deficiency [64]. Par-

aseptal emphysema involves destruction predominantly in peripheral subpleural lobules and is associated with male sex and higher incidence of spontaneous pneumothorax while less strongly associated with cigarette smoking, symptoms, and spirometric obstruction. Some COPD patients have bullae, defined as air-filled spaces exceeding 1 cm in diameter, which are often paraseptal in location and may enlarge to become giant bullae if they occupy at least one third of the hemithorax [65]. Bullectomy has long been performed in hyperinflated patients with giant bullae with good clinical outcomes [66].

Emphysema burden may be homogeneously similar across all lobes or heterogeneous if predominantly affecting the upper lobes (more common in smoking-related emphysema) or lower lobes (associated with AAT deficiency). Heterogeneous emphysema has often been defined by a $\geq 15\%$ difference in interlobar average voxel density on quantitative computed tomography (CT) (see Recommendation 10 for more details).

Most major BLVR trials excluded patients with giant bullae [14, 15, 31, 47, 48] or emphysema with homogeneous distribution, either by visual estimation [27, 32] or quantitative CT (online suppl. Table S1) [14, 15, 31, 33, 48]. However, the IMPACT trial exclusively enrolled patients with homogeneous disease ($<15\%$ discrepancy by quantitative CT) with BLVR outcomes similar to trials of heterogeneous disease [47]. Notably, IMPACT required a higher residual volume ($>200\%$) compared to trials of heterogeneous disease ($>150\text{--}180\%$) [14, 15, 27, 31–33, 47, 48]. No major trials to date specifically included or excluded participants based on emphysema subtype other than bullous disease. There are case reports and small series of giant bullae treated with endobronchial valves, with resolution of the bullae reported in some [67, 68].

Rationale for Recommendations

Major BLVR trials have included and demonstrated benefit in patients with heterogeneous and homogeneous emphysema, though a higher residual volume threshold ($>200\%$) was used in trials of homogeneous disease. BLVR should, therefore, be considered in patients regardless of lobar emphysema distribution, but a higher residual volume may be required in those with homogeneous disease.

Patients with giant bullae should be evaluated for bullectomy. Case reports of successful resolution of giant bullae after endobronchial valve placement in collateral

ventilation negative patients have been reported, though outcome data are limited, so BLVR in this setting should be limited to highly selected patients who are not operative candidates [67, 68]. Patients with predominantly paraseptal emphysema have been deemed poor candidates for BLVR based on expert opinion, though no major trials explicitly excluded such patients [7, 26]. Rationales for not offering BLVR in paraseptal-predominant disease include concern for increased pneumothorax risk, possible association of paraseptal disease with ipsilateral compensatory hyperinflation which limits the benefits of successful lobar atelectasis, and lack of perfusion or ventilation of paraseptal disease making them less likely to develop atelectasis after valve placement [7, 26, 69]. Many patients have elements of centrilobular and paraseptal emphysema; such patients should not be excluded from BLVR consideration unless paraseptal disease predominates.

Recommendation 8

- In patients with advanced, hyperinflated emphysema being considered for BLVR, we suggest avoiding valve placement ipsilateral to prior lobectomy or prior pleurodesis. Valve placement may be considered in the setting of prior contralateral lobectomy or pleurodesis, prior ipsilateral sub-lobar resection, or after sternotomy.

Literature Review

Most major BLVR trials excluded patients with prior thoracic surgery, variably including lung transplant, bullectomy, LVRS, and/or lobectomy (online suppl. Table S1) [14, 15, 27, 31, 47, 48]. Two trials additionally excluded those with prior sternotomy [31, 48]. Surgical interventions in the thorax promote pleural adhesion formation, hypothesized by some to be a risk factor for post-BLVR pneumothorax. Additionally, in the case of a prior ipsilateral lobectomy or bullectomy, additional volume loss in that hemithorax (to a functional pneumonectomy for the former) is likely to have different physiological outcomes than BLVR in a non-instrumented hemithorax. Two retrospective studies of patients treated in BLVR trials have assessed an association between suspected pleural adhesions visible on CT with periprocedure pneumothorax; one including 64 patients found a higher median number of adhesions in those who developed pneumothorax, while the other including 129 patients found adhesions were associated with a lower risk of pneumothorax (odds ratio: 0.33) [70, 71].

Rationale for Recommendations

Existing data regarding suspected pleural adhesions visible on pre-procedure CT scan and periprocedural pneumothorax are conflicting, and there is no other evidence of an association between peri-BLVR pneumothorax and prior sub-lobar resection, contralateral lobectomy, sternotomy, or lung transplant. The panel recommends against implanting valves in a lobe ipsilateral to a prior lobectomy or bullectomy as, in most cases, this would entail forcing atelectasis upon all or most residual lung parenchyma in that hemithorax.

No BLVR trials specifically excluded patients with prior pleurodesis. Pleurodesis would prohibit the medial shifting of an atelectatic treated lobe, which in theory might limit volume loss and/or increase risk of complications from persistent tension of the atelectatic lung pulling on the fused pleural surface. However, this is an area requiring further study to better understand outcomes in patients with prior pleurodesis.

Recommendation 9

- In patients with advanced, hyperinflated emphysema being considered for BLVR, we recommend against excluding patients based on AAT deficiency.

Literature Review

AAT-deficient emphysema patients were excluded from most clinical trials on BLVR. Three studies have addressed BLVR in AAT-deficient emphysema patients. Everaerts and colleagues retrospectively analyzed a cohort of 53 AAT-deficient patients who underwent BLVR, noting similar outcomes and complication rates as those reported by clinical trials of non-AAT-deficient patients [72]. Hillerdal and Mindus [73] reported a series of 15 AAT-deficient patients who underwent BLVR, and similarly noted acceptable clinical outcomes and no deterioration of lung function post-BLVR over 4 years of follow-up. Hogarth and colleagues [74] reported on a cohort of AAT-deficient patients enrolled in the multicenter EMPROVE trial and found outcomes were similar to the overall trial cohort.

Rationale for Recommendations

Existing BLVR data, including a substudy of a multicenter trial, suggest BLVR produces similar benefits in patients with severe emphysema with and without AAT-deficiency. We suggest that AAT-deficient emphysema patients should not be excluded based on this factor alone. Medical therapy should be optimized for all

prospective BLVR candidates; for AAT-deficient patients, this would include indicated antitrypsin augmentation via regular infusions prior to proceeding with BLVR in countries where this medication is available.

Recommendation 10

- In patients with advanced, hyperinflated emphysema being considered for BLVR, we suggest using quantitative CT analysis to estimate fissure integrity, extent of lobar emphysematous destruction, and lobar volumes in all patients to inform procedure candidacy and lobe targeting. We suggest lobes with extensive emphysema ($\geq 50\%$ low attenuation score at -910 Hounsfield units (HU) or $\geq 20\%$ at -950 HU) adjacent to fissures that are at least 80% intact be considered potential targets for valve placement.

Remarks: a densitometry percent low attenuation score threshold of $\geq 40\%$ at -920 HU has also been used in BLVR RCTs to define lobes eligible for treatment with valves and may also be considered. Lobes adjacent to fissures $<80\%$ intact are highly unlikely to develop atelectasis after valve treatment due to collateral ventilation and should be avoided. Lobes adjacent to fissures that are at least 80% intact should be subject to additional considerations before concluding a lobe is appropriate for valve implantation; see recommendations #7, 10, 11, 12, and 16 for other considerations regarding lobe selection for treatment.

Literature Review

Interobserver agreement on which lobe has the highest emphysema burden is only fair to moderate and improves when quantitative CT data are provided [75]. Most BLVR trials utilized quantitative CT analysis to assist with lobe targeting decisions, define between-lobe heterogeneity, or both [14, 15, 27, 31, 33, 47, 48]. Quantitative CT densitometry quantifies the burden of low attenuation emphysematous change by reporting the percentage of lobar parenchyma exceeding a specific HU threshold. Most trials using densitometry to define heterogeneity used >10 – 15% difference between ipsilateral lobes at -910 HU [15, 31, 47, 48]. Most trials using densitometry to determine which lobes had sufficient emphysema burden to be treated used $\geq 50\%$ destruction at -910 HU, though $\geq 40\%$ at -920 HU was also used. Historically, a threshold of -950 HU to define emphysematous destruction predominates in literature comparing quantitative CT to pathological findings or physiological assessments; in comparative studies utilizing CT series with <2 mm slice thickness, a -950 HU threshold correlates better with pathological emphysema burden than -910 HU [76–80].

Fissure completeness-related valve eligibility was informed by visual estimate in some trials [14, 27, 31, 32] and quantitative CT in others [15, 33, 47, 48], with most requiring $\geq 90\%$ intactness by either technique. In a retrospective study correlating quantitative CT-derived fissure completeness with detection of collateral ventilation by bronchoscopic balloon occlusion testing, collateral ventilation was always present below 90% completeness on the right and 80% completeness on the left, with lower likelihood of collateral ventilation as fissure integrity increased [81]. Another retrospective study found volume loss began to occur on post-procedure CT as fissure intactness increased above 80% [82].

Rationale for Recommendations

Quantitative CT assessment has been used to identify lobes with sufficient emphysema to warrant treatment, define between-lobe heterogeneity, and assess fissure completeness, all of which inform BLVR candidacy and optimal lobe targeting. Quantitative analysis is favored over visual estimates, given only fair to moderate agreement between expert readers on the aforementioned metrics in the absence of quantitative CT data. Precise destruction thresholds at which patients might benefit from a lobe being treated have not been well elucidated. While most trials used a threshold of $\geq 50\%$ destruction at -910 HU, the literature correlating emphysema burden to pathological and physiological metrics suggests -950 HU more accurately reflects emphysema burden when a thin cut (<2 mm) CT series is analyzed. The scenario of sufficient ($\geq 20\%$) destruction at -950 HU while emphysema burden at -910 HU is under 50% suggests a high degree of intralobar heterogeneity, which is common in predominantly paraseptal emphysema, which is addressed in Recommendation 7. A fissure completeness threshold of 80% has been associated with lack of collateral ventilation and post-procedure volume loss in some patients, though existing data are limited and a higher threshold of 90% was identified as required in the right lung in one study which considered left and right separately. Higher completeness correlates with lower odds of collateral ventilation being present.

Recommendation 11

- In patients with advanced, hyperinflated emphysema being considered for BLVR who have multiple potential lobar targets or a homogeneous disease distribution, we suggest consideration of pre-procedure perfusion testing.

Remarks: in the absence of more robust data confirming the impact of pre-procedure perfusion

quantification on patient-centered outcomes, perfusion assessment is considered optional and may be omitted if not readily or locally available.

Literature Review

Among BLVR trials, IMPACT assessed perfusion in all patients and specifically targeted the lobe with the least perfusion; this trial exclusively enrolled patients with homogeneous disease distribution [47]. Two others enrolling patients with heterogeneous distribution obtained perfusion studies in at least some patients, with perfusion playing a role in lobe targeting decisions if there were multiple targets with similar emphysema burden [14, 48].

A few studies have evaluated the contribution of perfusion assessment on BLVR outcomes. An analysis of the treatment group in the VENT trial found those with low pre-procedure perfusion in the treated lobe had greater post-procedure improvement in 6-min walk distance when compared to those in whom more highly perfused lobes were treated [31, 83]. Another study found target lobe perfusion was not associated BLVR outcomes, though a high ipsilateral nontarget lobe perfusion was associated with an increase in post-procedural 6-min walk distance [84]. Ide Bolet and colleagues [85], in a prospective observational study of 92 BLVR candidates with homogeneous emphysema, used single-photon emission CT to assess lobar perfusion to inform lobe targeting; favorable patient outcomes were reported, but no comparison was made to an alternative lobe targeting strategy.

Other studies described various technologies to assess lung perfusion prior to BLVR, including perfusion scintigraphy, single-photon emission computed tomography, and dual-energy contrast-enhanced chest CT scan, and found them to be equivalent in prediction of perfusion [86–89]. Single-photon emission computed tomography and dual-energy contrast-enhanced chest CT scan have the potential advantage of providing more precise estimates of lobar perfusion.

Rationale for Recommendations

The optimal lobar BLVR target has the highest degree of emphysematous destruction, which is typically accompanied by a corresponding reduction in ventilation (V) and perfusion (Q). Inadvertent treatment of a lobe with relatively high perfusion could contribute to significant V/Q mismatch post-procedure, which could attenuate clinical benefit or increase risk. How well quantitative CT assessment of emphysema burden correlates with perfusion, and

which is more clinically meaningful, has not been described.

To date, there is no robust evidence that universal pre-procedure perfusion testing is associated with improved BLVR outcomes. However, there is some evidence that treatment of lobes with less baseline perfusion is associated with higher magnitudes of benefit, and the only RCT involving only patients with homogeneous disease specifically targeted the least perfused lobe. In this context, pre-BLVR perfusion testing may be beneficial when there are multiple lobar targets (e.g., upper-lobe predominant disease with highly intact fissures bilaterally) or homogeneously distributed disease. Detecting a particularly underperfused lobe in these scenarios would make it the evidence-based preferred target.

Recommendation 12

- In patients with advanced, hyperinflated emphysema being considered for BLVR, targeting lobe(s) in a lung with significantly lower overall emphysema scores or greater perfusion should be avoided.

Literature Review

There are limited direct data regarding BLVR outcomes stratified by whether valves were placed in the overall more diseased lung or not. Among major BLVR trials, several included contingencies related to overall lung function or emphysema burden. IMPACT, which included only patients with homogeneous disease distribution, excluded patients with left vs. right discrepancy in perfusion of 20% or greater [47]. LIBERATE excluded patients with $\geq 75\%$ destruction (assessed at -910 HU) in all lobes in either lung [15]. Several other trials discuss lung targeting independently of lobe targeting, but only to state that the lung with maximal interlobar heterogeneity dictated which lung to treat [15, 31]. There is limited detail in the existing literature about BLVR-related mortality. The LIBERATE trial noted that three deaths related to pneumothorax all occurred in patients in whom the most diseased lobe was not treated as the protocol emphasized treating the lung with maximal heterogeneity, which likely meant the healthier lung was treated in these cases [15].

Rationale for Recommendations

Patients undergoing BLVR are at significant risk of sudden major clinical decompensation, most notably from pneumothorax, which is common and occurs almost exclusively on the side in which valves were placed. Patients with advanced emphysema complicated by

hyperinflation significant enough to warrant consideration of BLVR also tend to have limited physiological reserve. In patients with one lung clearly more diseased than the other, valves placed into the healthier lung complicated by pneumothorax could, therefore, result in a suddenly dire physiological situation. Therefore, the panel suggests that if there is a notable right versus left lung discrepancy in overall emphysematous burden assessed by quantitative CT or whole lung perfusion, one should avoid treating a lobe in the lung with less emphysema or with greater perfusion. The degree of right versus left discrepancy with either modality associated with increasing risk has not been well defined, however. The only trial to require pre-BLVR perfusion, IMPACT, excluded those with right-left perfusion differential of >20%.

Recommendation 13

- In patients with advanced, hyperinflated emphysema being considered for BLVR who have multiple potential lobar targets (see statement #10), we suggest using emphysema destruction scores, lobe volume, lobar perfusion, and presence of disqualifying incidental findings to determine the primary target lobe.

Remarks: qualitative inspection of the CT scan must be performed. Valves should not be routinely placed in lobes with large bulla, active infectious processes, or those with abnormalities requiring follow-up imaging, such as indeterminate lung nodules.

Literature Review

Recommendation 10 describes identification of lobes which might be eligible for valve implantation based on emphysema burden and completeness of adjacent fissures. In some cases, there may be multiple potential targets. Among BLVR trials, primary lobe selection in this scenario was variably driven by maximal interlobar heterogeneity [15, 31], maximal emphysema burden [14, 32, 48], the lowest perfusion [47], and maximal fissure completeness [33], with maximized emphysema burden considered secondarily in most trials where this was not the primary factor (online suppl. Table S1).

Recommendation 11 reviews the impact of perfusion data on lobe selection and BLVR outcomes. An analysis of prospectively collected multicenter registry data involving 236 patients assessed the impact of interlobar heterogeneity on clinical outcomes, noting the degree of heterogeneity did not predict meaningful clinical response (which was observed in most) [90]. Radiologic lung volume parameters, such as lobar vital capacity,

lobar total capacity, and the ratio of the two, may have benefit, as shown in a small retrospective study in which the lobe with the greatest deflation (lobar vital capacity/lobar total capacity) weakly correlated with the optimal target lobe [91]. Most BLVR trials excluded patients with recent active infection or indeterminate lung nodules within the target lobe, to allow optimal treatment and prevent masking of a developing problem like an early lung cancer [14, 15, 31, 47, 48].

Rationale for Recommendations

Both quantitative and qualitative assessment of CT scans should occur to optimize selection of the target lobe. Qualitative analysis should focus on discovery of large blebs or predominantly paraseptal emphysema, assessment of active areas of inflammation or infection, or indeterminate pulmonary nodules. Each of these conditions should be addressed prior to consideration of BLVR. Quantitative analysis producing emphysema destruction scores, perfusion evaluation, and approximation of lobe volume should be undertaken once the patient does not have evidence of a qualitative disqualifier.

Summarizing recommendations 7, 8, 10–13: the optimal BLVR target lobe has the highest emphysema burden, lowest perfusion, maximal heterogeneity compared to the adjacent ipsilateral lobe(s), intact adjacent fissures, lobar volume similar or higher than adjacent ipsilateral lobes, resides in a lung which has not been subjected to prior upper or lower lobectomy or pleurodesis and is overall more or similarly diseased as the contralateral lung, and lacks features such as large bullae, predominantly paraseptal emphysema, active infection, or indeterminate lung nodules.

Recommendation 14

- In patients with advanced, hyperinflated emphysema being considered for BLVR, we recommend against excluding patients based on a perceived increased risk of post-procedure pneumothorax.

Literature Review

The VENT trial in 2010 showed a low incidence of pneumothorax periprocedurally in patients randomized to obtain EBV, but this study did not select patients based on fissure completeness and lack of collateral ventilation [31]. Subsequent trials, most of which enhanced selection of collateral ventilation negative patients via bronchoscopic balloon occlusion testing prior to valve placement, reported higher rates of

periprocedural pneumothorax ranging 12–29% [14, 15, 27, 32, 47, 48]. A 2017 meta-analysis showed a relative risk of pneumothorax in the EBV group versus control of 9.65 ($p = 0.0001$) [92]. A population-level analysis of BLVR-related admissions in the USA estimated an incidence of pneumothorax of 26% in real-world practice [93].

High fissure integrity, which is highly associated with absence of collateral ventilation and subsequent volume loss, has been reliably associated with pneumothorax, which occurs as a consequence of shifts resulting from volume loss [94, 95]. Several other potential risk factors for pneumothorax have been hypothesized, including radiographically evident pleuro-parenchymal adhesions (also discussed in Recommendation 8), significant paraseptal emphysema, significant volume discrepancy compared to adjacent untreated lobe(s), high degree of emphysematous destruction in the adjacent untreated lobe(s), treatment of an upper lobe, and rapid onset of atelectasis [95–98]. Among these, treatment of an upper lobe has been most reliably associated with pneumothorax, with up to a 40% incidence when treating the left upper lobe [98]. For other potential risk factors, associations with pneumothorax are methodologically limited at present.

Rationale for Recommendations

Post-BLVR pneumothorax is common and no studies to date have determined reliable predictors of its occurrence, other than highly intact fissures which are necessary for the mechanism of action of valve-associated volume loss and, therefore, a desirable attribute. While accruing evidence suggests treatment of an upper lobe may confer more risk of pneumothorax, studies remain methodologically limited. In this context, all patients selected for this procedure should be judged physiologically capable of tolerating a pneumothorax, and physiologically frail patients should not be offered BLVR because their risk of pneumothorax is perceived to be low.

Procedural Recommendations

Recommendation 15

- In patients with advanced, hyperinflated emphysema undergoing bronchoscopy for BLVR, we suggest performing a direct physiological assessment for collateral ventilation (e.g., bronchoscopic balloon occlusion) before placing valves in the left lung if fissure integrity score is <95% and prior to all right-sided valve placements.

Remarks: collateral ventilation may still be present in a minority of those with a highly intact (>95%) left major

fissure, making a direct physiological assessment prior to all left-sided valve placements a reasonable choice as well. It is also reasonable to place valves if there is concern about the quality or accuracy of a direct physiological assessment in the context of a highly intact fissure in either lung.

Literature Review

Successful BLVR requires absence of collateral ventilation through fissures bordering the target lobe. This can be assessed radiographically with quantitative or qualitative fissure completeness assessments or physiologically by occluding a potential target lobe during bronchoscopy with a balloon catheter then measuring the flow of air through the catheter tip extending distal to the balloon to directly detect the presence of collateral ventilation [99]. Most early BLVR trials utilized visual assessment of fissure integrity to determine eligibility for valves [27, 31]. Use of direct physiological collateral ventilation assessment via balloon occlusion was introduced in 2013 in an observational study demonstrating significantly improved likelihood of significant volume loss when a lobe with absent collateral ventilation by balloon occlusion testing was treated (58%) versus lobes with detectable collateral ventilation (8%) [99]. Most subsequent clinical trials required absence of collateral ventilation by bronchoscopic balloon occlusion testing prior to valve implantation [15, 27, 47, 48].

Two observational studies have evaluated the relationship between quantitative CT fissure assessment and bronchoscopic balloon occlusion detection of collateral ventilation. A multicenter observational study involving 240 BLVR patients with pre-procedure quantitative CT analysis and a balloon occlusion physiological assessment noted increasing probability of collateral ventilation absence as fissure integrity scores increased and identified a left-sided minimal fissure score of 80% and right-sided minimal score of 90%, below which all assessed fissures demonstrated collateral ventilation [81]. Left major fissure scores exceeding 95% were associated with collateral ventilation in fewer than 10%. A smaller study of 34 patients also evaluated the association of post-BLVR volume loss and clinical outcomes with fissure integrity scores and balloon occlusion assessment, noting both modalities similarly predicted clinical response [100].

Rationale for Recommendations

Direct physiological assessment via bronchoscopic balloon occlusion testing has demonstrated a good ability to predict lobes which will become atelectatic after

lobar occlusion with valves. Relying on quantitative CT alone will result in a significant minority of patients undergoing futile valve placement, leading to unnecessary hospital admission and an additional procedure to remove ineffective valves. Direct physiological assessment is fast, adds little if any risk, and prevents valve placement into lobes with significant collateral ventilation, often allowing a secondary target lobe to be successfully treated instead. Additionally, proprietary quantitative CT analysis algorithms may be subject to change over time, favoring direct assessments.

However, direct physiological assessment via balloon occlusion has several known issues, including early airway collapse, low flow states, repeated mucous plugging, and anatomical concerns, which can lead to inconclusive assessments in which collateral ventilation is not detected, or false-positive detection when none is present. The panelists suggest that in patients with CT evidence of a highly intact fissure but inconclusive direct physiological assessment, valve placement can be considered on a case-by-case basis, weighing the risks and benefits when a false-positive collateral ventilation assessment is suspected.

Recommendation 16

- In patients with advanced, hyperinflated emphysema who undergo BLVR, we recommend post-procedure hospital admission for a minimum of three nights.

Literature Review

Major BLVR trials reported significant rates of serious periprocedural complications, most notably pneumothorax, reviewed in detail in Recommendation 14. In most trials, 66–79% of post-BLVR pneumothoraces occurred within 3 days of the procedure and most required chest tube thoracostomy [14, 15, 33, 71]. In this patient population with limited physiological reserves, pneumothorax can be rapidly life-threatening and often needs immediate intervention, making it difficult and risky to manage in an ambulatory setting. In LIBERATE, three of four periprocedural deaths in the treatment arm involved pneumothorax [15]. In the USA, the Food and Drug Administration-approved Instructions for Use document for one endobronchial valve notes a requirement for post-BLVR hospital admission for at least three nights [101].

Rationale for Recommendations

Post-BLVR complications are common, most notably pneumothorax, which usually occurs within 3 days of valve placement and has been associated with mortality in the periprocedural period. Patients meeting criteria for BLVR have limited pulmonary reserve and are consequently less tolerant of serious complications. As reviewed in Recommendation 14, reliable predictors of periprocedural pneumothorax have not been well described. In this context, the panel recommends hospital admission for at least three nights following BLVR.

Post-Procedure Management

Recommendation 17

- In patients with advanced, hyperinflated emphysema who have undergone BLVR, we suggest patients be evaluated for clinical/functional status, radiographic response, and complications at 4–6 weeks, 3–6 months, and 12 months after procedure.

Literature Review

Published data regarding post-BLVR follow-up consist of experts describing practices at their centers, and several reports relate to valve revision for treatment non-response or loss of response [26, 69, 102]. These publications suggest close initial follow-up for assessment of adverse events and success of the procedure. Post-procedure CT scans delineate degree of atelectatic response to valves and evaluate for valve malposition, and physiologic testing such as pulmonary function testing and 6-min walk distance are commonly used to objectively assess physiological response. In the limited literature on valve revisions, most for loss of initial clinical benefit, bronchoscopy was performed in 20–41% of cases at a median of 200–360 days [102, 103].

Rationale for Recommendations

Optimal follow-up intervals and testing are not well defined in the current literature. Follow-up is important to identify complications and assess for atelectatic and clinical response to valve placement, as well as later loss of clinical improvement or radiographic atelectasis. The panel suggests the aforementioned follow-up cadence, including follow-up through at least 1 year given median revisions reported in the literature in the 6–12 months post-BLVR. Physiological assessments, including PFTs and 6-min walk distance, and CT scans to assess for post-procedure atelectasis and valve positions can inform post-procedure care.

Recommendation 18

- In patients with advanced, hyperinflated emphysema with subtotal atelectasis of the target lobe without clinical improvement by 3 months post-BLVR, we suggest evaluation with CT chest and consideration of bronchoscopic inspection with valve revision or removal if indicated.

Literature Review

Some BLVR patients do not develop an atelectatic response to valve placement, which may be due to the presence of collateral ventilation not detected prior to placement, valve malposition, valve malfunction, valve undersizing or bronchial stretching with perivalvular leak, or perivalvular granulation tissue [26, 102]. There are several reports of valve revision which include patients who never achieved initial radiographic atelectasis. In a series of 179 BLVR patients, 74 (41%) required at least one revision, including 23 (13%) who never developed atelectasis after initial BLVR [102]. Thirteen of these 23 patients were found to have potentially correctable reasons for non-response at revision (e.g., significant perivalvular granulation tissue, valve malposition), though it is not reported how many achieved atelectasis after revision. In another revision series, 30 of 148 patients required at least one valve revision, nine for lack of initial response, two of whom achieved atelectasis after revision [103]. Another series of 244 patients reported 73 patients underwent at least one revision bronchoscopy, most commonly for lack of clinical response [104]. Post-revision, 36% of patients developed atelectasis, though because some patients were revised for reasons other than lack of initial response, it is not clear how many initial non-responders developed atelectasis.

Rationale for Recommendations

Available evidence suggests patients should experience meaningful clinical and radiographic benefit from BLVR by 3 months post-procedure, with radiographic atelectasis typically occurring within weeks of the procedure. If no radiographic atelectasis is apparent, evaluation of valve position with chest CT and revision bronchoscopy are indicated. Chest CT may identify valves which are malpositioned or undersized, or presence of untreated segments. Chest CT is not sensitive for perivalvular granulation tissue or valve malfunction, which are typically detected via revision bronchoscopy. In the absence of any of these findings,

presence of occult collateral ventilation is more likely, and reassessment via balloon occlusion can be considered.

Recommendation 19

- In patients with advanced, hyperinflated emphysema who develop complete atelectasis of the target lobe without clinical improvement by 3 months post-BLVR, we suggest PFTs, CT chest, cardiac evaluation, and other testing guided by clinical evaluation.

Literature Review

Some patients achieve complete target lobe atelectasis or exceed the minimal clinically important difference for FEV1 after BLVR but do not draw subjective benefit [105, 106]. This has been attributed to compensatory hyperinflation of the nontreated adjacent lobe, comorbid lung disease (e.g., fibrotic lung disease), cardiac comorbidities, valve complications, or other anatomical considerations like airway kinking [40, 105, 107, 108]. Compensatory hyperinflation is a post-BLVR phenomenon in which significant atelectasis of the target lobe occurs but is accompanied by increased hyperinflation of the ipsilateral lobe(s), resulting in overall attenuation or obviation of volume loss and failure to improve diaphragmatic mechanics, with precise post-BLVR incidence unknown [106].

Rationale for Recommendations

Patients who appropriately achieve atelectasis but do not improve clinically should be assessed for an alternative reason for a lack of improvement. Post-procedure quantitative CT lung volumes may detect compensatory hyperinflation of the ipsilateral untreated lobe(s). There is also significant interplay between intrathoracic volumes and hemodynamic physiology that should be assessed with a cardiac evaluation, and cardiac comorbidities are relatively common in patients with a history of cigarette smoking. Other possible testing includes evaluation for pulmonary emboli, obstructive sleep apnea, and other etiologies based on clinical suspicion.

Recommendation 20

- In patients with advanced, hyperinflated emphysema who experience a clinical and radiographic response post-BLVR followed by loss of clinical and radiographic response, we

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(continued)

suggest CT chest and revision bronchoscopy to evaluate for valve malposition or malfunction, granulation tissue, infection, or other etiology.

Literature Review

Valve revision studies discussed in Recommendation 18 also included patients achieving initial atelectatic response to valve placement who lost atelectasis at a later date, often accompanied by loss of clinical improvement [102, 103]. In a series of 179 patients, 32 of 74 patients who underwent revision did so due to loss of effect [102]. Valve-associated granulation tissue was noted in over half of revisions, particularly in loss-of-effect cases, often managed with valve removal with later replacement. About two-thirds of patients undergoing revision bronchoscopy achieved higher FEV1 or lower RV after revision. In another series of 148 BLVR, 34 of 43 revisions were due to loss of initial effect, with 24 (71%) regaining atelectasis after revision [103].

Rationale for Recommendations

There are limited data on valve revision, though in existing series, most patients with an initial atelectatic response to valves regain response after revision, which underlies this recommendation to pursue revision bronchoscopy guided in part by pre-revision chest CT scan in this scenario. Additionally, other etiologies such as airway folding, post-obstructive pneumonia, worsening of concomitant pulmonary disease, and disease in other organ systems should be considered on a case-by-case basis. Further, initial informed discussion with the patient should describe the possible need for further bronchoscopic intervention prior to initial procedural intervention.

Recommendation 21

- In patients with advanced, hyperinflated emphysema who experience a marked increase in supplemental oxygen requirement in the perioperative period, we suggest evaluation for reversible etiologies, such as pneumothorax, pneumonia, or pulmonary emboli.

Remarks: mild-to-moderate escalation of supplemental oxygen requirement is common post-procedure and usually transient; if it persists longer than 4–6 weeks, we suggest additional cardiopulmonary evaluation.

Literature Review

Among major BLVR trials, LIBERATE reported post-BLVR hypoxemia in 5.5% of treated patients; others reported lower rates or did not specifically report on hypoxemia or respiratory failure [15, 31, 48]. One single center retrospective study specifically reported on the impact of BLVR on gas exchange in 77 patients over 6 months post-procedure [109]. Fifty-six percent required supplemental oxygen at baseline pre-procedure. Mean alveolar-arterial oxygen gradient was higher at 1-month post-procedure than baseline but decreased back to baseline at 3 and 6 months post-procedure. Functional status was improved at 1-month post-procedure despite the larger A-a gradient, and while most patients remained on similar oxygen supplementation as pre-procedure, there was less exertional desaturation by 6 months post-procedure. The main purported mechanism of transient BLVR perioperative hypoxemia is V/Q mismatch until residual perfusion to the atelectatic treated lobe decreases [109]. A population-level analysis of BLVR-related admissions in the USA estimated an incidence of acute respiratory failure of 19.5%, with 12% of those requiring mechanical ventilation, though this study did not discriminate between hypoxemic and hypercapnic respiratory failure [93].

Other conditions which commonly cause hypoxemia were also reported in BLVR trials, including pneumothorax, COPD exacerbations, and pneumonias, typically occurring in under 5% of treated patients [14, 15, 27, 47, 48]. Pulmonary embolism has been described in case reports in the BLVR perioperative period [110].

Rationale for Recommendations

Transient mild-moderate hypoxemia appears common in the immediate post-BLVR period, likely due to transient V/Q mismatch, but is typically resolved by 3 months post-procedure. Severe post-procedure hypoxemia should prompt evaluation for additional contributing etiologies, including pneumothorax, COPD exacerbation, pneumonia, venous thromboembolism, pulmonary edema, or right-to-left intracardiac shunt exacerbated by increased post-operative pulmonary vascular resistance.

Recommendations failing Consensus or with Insufficient Evidence to Support a Recommendation

The panel was unable to reach consensus related to pre- or post-BLVR pulmonary rehabilitation. Pre-procedure pulmonary rehabilitation was required for inclusion in two major BLVR trials and required for some in a third [14, 15, 31]. A subsequent multicenter

RCT in which patients were randomized to rehabilitation before BLVR, rehabilitation after BLVR, or no rehabilitation, found no difference in post-procedure endurance between these groups [111]. Additionally, a post hoc analysis of the VENT trial found that pre-BLVR pulmonary rehab increased the 6-min walk distance by a mean of 14 m, short of the minimum clinically important difference of 26 m, though patients unable to work more than 250 m at baseline were more likely to improve beyond the minimum clinically important difference [112].

Despite lacking clear evidence that pulmonary rehabilitation impacts post-BLVR endurance, some panel members advocated for its routine use given it is a safe intervention shown to routinely benefit patients with advanced COPD, is recommended by Global Initiative for Chronic Obstructive Disease guidelines, and BLVR involves hospitalization which promotes deconditioning [113, 114]. Additionally, several panel members routinely utilize rehabilitation both before and after BLVR, a strategy which has not yet been compared to a no-rehabilitation strategy. Other panel members noted that access to pulmonary rehabilitation may be limited due to proximity, capacity, transportation, insurance, and/or financial constraints for many; therefore, a rehabilitation requirement (particularly without clear evidence of benefit) would be overly onerous.

The panel agreed that existing evidence is insufficient to make a recommendation for or against sequential bilateral BLVR, which has been the subject of three retrospective studies with total $n = 39$ [115–117]. Treatment of the contralateral lung was pursued due to loss of benefit from index BLVR in one (median 18 months later) and upfront planned sequential bilateral BLVR in two (median 128–199 days after index BLVR). Clinical benefit was reported in two of three, though no functional improvement was noted in one despite a significant mean increase in FEV1 and decrease in residual volume after the contralateral procedure. Complication rates were higher in sequential bilateral BLVR compared to unilateral BLVR in two studies. Larger, more robust studies are necessary to determine the safety, efficacy, and appropriate patient selection before advocating for routine bilateral sequential BLVR.

The panel agreed that existing evidence is insufficient to make a recommendation for or against placing valves in a staged fashion over two procedures instead of placing all in a single procedure. A 2012 trial randomized 73 patients to sham bronchoscopy vs. bilateral subtotal upper lobar occlusion with valves (leaving one segment untreated in the right upper lobe and the lingula) [118].

Despite subtotal occlusion, 24% of treated patients noted improved functional status and there was mean lobar volume loss of 9% with no pneumothoraces, which are commonly attributed to significant or rapid volume shifts within the hemithorax. Based on these observations, staged BLVR has been hypothesized to decrease the risk of pneumothorax by shifting volume in smaller increments across two procedures. Two multicenter retrospective studies of consecutively treated patients treated in a staged fashion have reported incidences of post-BLVR pneumothorax of 7% and 12%, significantly lower than those noted in most major BLVR trials, with comparable clinical responses [119, 120]. However, no comparative studies have been performed and there are significant potential limitations to staged BLVR, including longer cumulative hospital admission and greater cost. Prospective randomized trials of staged vs. nonstaged BLVR are required. While awaiting these trials, noting signals toward increased overall safety, staged BLVR may be a consideration in patients of particularly tenuous physiological status.

A single multicenter retrospective before-and-after cohort study has associated use of low intraprocedural FiO₂ (mean 0.29) to a reduced risk of post-BLVR PTX (7%) compared to high intraprocedural FiO₂ (mean 0.95, PTX in 31%) [121]. Pneumothoraces in the high FiO₂ group also occurred more rapidly (6 h vs. 81 h post-procedure). The hypothesized mechanism of benefit is slower absorption atelectasis in treated lobes not washed of nitrogen by a high oxygen concentration. While this intervention appears safe in the highly monitored intraoperative setting in which it was applied, significant methodological issues limit confidence in the validity of this approach which requires further study.

Conclusion

BLVR represents an important therapeutic option for patients with advanced, hyperinflated emphysema, in whom BLVR can lead to significant improvements in pulmonary function, exercise capacity, and quality of life. Careful patient selection is paramount for optimizing outcomes, which is optimally informed by clinical history, examination, quantitative CT scan analysis, direct physiological assessment for collateral ventilation, and perfusion studies. Further research will continue to clarify the role these data play in selecting optimal patients and optimal target lobes. Many recommendations were informed by the inclusion and exclusion criteria of major trial dates, many of which were somewhat

arbitrarily chosen and may exclude some patients who would benefit from BLVR. Dedicated study of many reviewed topics to delineate more precise thresholds predicting benefit or risk will better inform future BLVR medical decision-making. The goal of these expert recommendations is to support BLVR patients in the peri and post-procedural setting and achieve meaningful quality of life improvements in this patient population with severe COPD.

Statement of Ethics

This study involved a review of the literature; no research was performed on human subjects.

Conflict of Interest Statement

C.S. has received research funding to MUSC from CSA Medical and NuVaira for devices in COPD. He is a consultant to Morair and Rhyme. G.J.C. has received research grants to Temple University from GalaxoSmithKline, Boehringer Ingelheim, Chiesi, Apreo, Morair, Pulmonx, Olympus, Gilead, Genentech, Bristol Myers Squibb, Sanofi, Regeneron, Ubiquity, and Gravitax and consultant fees from GalaxoSmithKline, Boehringer Ingelheim, Chiesi, Apreo, Morair, Pulmonx, Olympus, Gilead, Genentech, Bristol Myers Squibb, Sanofi, Regeneron, Ubiquity, and Gravitax. M.M.W. has received research funding from Pulmonx and consultant fees from Pulmonx and Olympus. No other

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Author Contributions

Study concept and design: M.M.W., F.J.H., G.J.C., and R.J.L. Acquisition of data: M.M.W., R.J.L., G.J.C., D.G., A.J.I., S.L., F.C.S., P.L.S., D.-J.S., C.S., and F.J.H. Analysis and interpretation of data: M.M.W. and R.J.L. Drafting of the manuscript: R.J.L., M.M.W., and C.M.K. All authors participated in critical revision of the manuscript for important intellectual content and provided final approval to submit this version of the manuscript and have agreed to be accountable for all aspects of the work. M.M.W. and R.J.L. had full authority over study design; data collection, management, analysis, and interpretation; writing of the report; and the decision to submit the report for publication

Data Availability Statement

Literature search and consensus voting files are in the possession of the primary authors and can be provided upon request.

References

- Adeloye D, Song P, Zhu Y, Campbell H, Sheikh A, Rudan I, et al. Global, regional, and national prevalence of, and risk factors for, chronic obstructive pulmonary disease (COPD) in 2019: a systematic review and modelling analysis. *Lancet Respir Med*. 2022;10(5):447–58. [https://doi.org/10.1016/S2213-2600\(21\)00511-7](https://doi.org/10.1016/S2213-2600(21)00511-7)
- Boers E, Barrett M, Su JG, Benjafield AV, Sinha S, Kaye L, et al. Global burden of chronic obstructive pulmonary disease through 2050. *JAMA Netw Open*. 2023;6(12):e2346598. <https://doi.org/10.1001/jamanetworkopen.2023.46598>
- Lechner AJ, Matuschak GM, Brink DS. Respiratory: an integrated approach to disease. McGraw-Hill Medical; 2012. [cited 2024 Nov 18]. Available from: <https://cir.nii.ac.jp/crid/1130282270386791424>
- Janson C, Marks G, Buist S, Gnatiuc L, Gislason T, McBurnie MA, et al. The impact of COPD on health status: findings from the BOLD study. *Eur Respir J*. 2013;42(6):1472–83. <https://doi.org/10.1183/09031936.00153712>
- Kim YW, Lee C-H, Hwang H-G, Kim Y-I, Kim DK, Oh Y-M, et al. Resting hyperinflation and emphysema on the clinical course of COPD. *Sci Rep*. 2019;9(1):3764. <https://doi.org/10.1038/s41598-019-40411-1>
- Dubé BP, Guerder A, Morelot-Panzini C, Laveneziana P. The clinical relevance of the emphysema-hyperinflated phenotype in COPD. *COPD Res Pract*. 2016;2(1):1. <https://doi.org/10.1186/s40749-015-0017-7>
- Shah PL, Herth FJ, van Geffen WH, Deslee G, Slebos D-J. Lung volume reduction for emphysema. *Lancet Respir Med*. 2017;5(2):147–56. [https://doi.org/10.1016/S2213-2600\(16\)30221-1](https://doi.org/10.1016/S2213-2600(16)30221-1)
- Aliverti A. Chest wall mechanics in COPD. *Curr Respir Med Rev*. 2008;4(4):240–9. <https://doi.org/10.2174/157339808786263815>
- Arnold MT, Dolezal BA, Cooper CB. Pulmonary rehabilitation for chronic obstructive pulmonary disease: highly effective but often overlooked. *Tuberc Respir Dis*. 2020;83(4):257–67. <https://doi.org/10.4046/trd.2020.0064>
- Agusti A, Vogelmeier CF. Gold 2024: a brief overview of key changes. *J Bras Pneumol*. 2023;49(6):e20230369. <https://doi.org/10.36416/1806-3756/e20230369>
- Fishman A, Martinez F, Naunheim K, Piantadosi S, Wise R, Ries A, et al. A randomized trial comparing lung-volume-reduction surgery with medical therapy for severe emphysema. *N Engl J Med*. 2003;348(21):2059–73. <https://doi.org/10.1056/NEJMoa030287>
- Todd JL, Palmer SM. Lung transplantation in advanced COPD: is it worth it? *Semin Respir Crit Care Med*. 2010;31(3):365–72. <https://doi.org/10.1055/s-0030-1254076>
- Hsu K, Williamson JP, Peters MJ, Ing AJ. Endoscopic lung volume reduction in COPD: improvements in gas transfer capacity are associated with improvements in ventilation and perfusion matching. *J Bronchology Interv Pulmonol*. 2018;25(1):48–53. <https://doi.org/10.1097/LBR.0000000000000445>

- 14 Criner GJ, Delage A, Voelker K, Hogarth DK, Majid A, Zgoda M, et al. Improving lung function in severe heterogeneous emphysema with the spiration® valve system (EMPROVE): a multicenter, open-label, randomized, controlled trial. *Am J Respir Crit Care Med*. 2019;200(11):1354–62. <https://doi.org/10.1164/rccm.201902-0383OC>
- 15 Criner GJ, Sue R, Wright S, Dransfield M, Rivas-Perez H, Wiese T, et al. A multicenter randomized controlled trial of zephyr endobronchial valve treatment in heterogeneous emphysema (LIBERATE). *Am J Respir Crit Care Med*. 2018;198(9):1151–64. <https://doi.org/10.1164/rccm.201803-0590OC>
- 16 Pietzsch JB, Busca R, Rott C, Geisler BP, Weber SA, Slebos D-J, et al. Adoption patterns of bronchoscopic lung volume reduction procedures in Germany and predicted procedure volumes for other European countries. *Respiration*. 2019;97(1):34–41. <https://doi.org/10.1159/000491677>
- 17 Harris PA, Taylor R, Thielke R, Payne J, Gonzalez N, Conde JG. Research electronic data capture (REDCap)—A metadata-driven methodology and workflow process for providing translational research informatics support. *J Biomed Inform*. 2009;42(2):377–81. <https://doi.org/10.1016/j.jbi.2008.08.010>
- 18 Harris PA, Taylor R, Minor BL, Elliott V, Fernandez M, O'Neal L, et al. The REDCap consortium: building an international community of software platform partners. *J Biomed Inform*. 2019;95:103208. <https://doi.org/10.1016/j.jbi.2019.103208>
- 19 Cottin V, Martinez FJ, Smith V, Walsh SLF. Multidisciplinary teams in the clinical care of fibrotic interstitial lung disease: current perspectives. *Eur Respir Rev*. 2022;31(165):220003. <https://doi.org/10.1183/16000617.0003-2022>
- 20 Heinke MY, Vinod SK. A review on the impact of lung cancer multidisciplinary care on patient outcomes. *Transl Lung Cancer Res*. 2020;9(4):1639–53. <https://doi.org/10.21037/tlcr.2019.11.03>
- 21 NICE. Overview endobronchial valve insertion to reduce lung volume in emphysema Guidance. NICE; 2017 [Internet] [cited 2024 Nov 18]. Available from: <https://www.nice.org.uk/guidance/ipg600>
- 22 Martin MJ, Duloheroy Scrodin MM, Edell ES, Aguirre EA, Rajagopalan S, Bartholmai BJ, et al. Bronchoscopic lung volume reduction: highlighting the patient selection process. *Mayo Clin Proc*. 2023;98(9):1347–52. <https://doi.org/10.1016/j.mayocp.2023.05.001>
- 23 Chew J, Mahadeva R. The role of a multidisciplinary severe chronic obstructive pulmonary disease hyperinflation service in patient selection for lung volume reduction. *J Thorac Dis*. 2018;10(Suppl 27):S3335–43. <https://doi.org/10.21037/jtd.2018.07.36>
- 24 Welling JBA, Hartman JE, Augustijn SWS, Kerstjens HAM, Vanfleteren LEGW, Klooster K, et al. Patient selection for bronchoscopic lung volume reduction. *Int J Chron Obstruct Pulmon Dis*. 2020;15:871–81. <https://doi.org/10.2147/COPD.S240848>
- 25 Sakaguchi T, Hartman JE, Slebos D-J. An update on endobronchial valve therapy for severe emphysema: real world data and special indications. *Expert Rev Respir Med*. 2024;18(12):1003–11. <https://doi.org/10.1080/17476348.2024.2431522>
- 26 Klooster K, Slebos D-J. Endobronchial valves for the treatment of advanced emphysema. *Chest*. 2021;159(5):1833–42. <https://doi.org/10.1016/j.chest.2020.12.007>
- 27 Klooster K, ten Hacken NHT, Hartman JE, Kerstjens HAM, van Rikxoort EM, Slebos D-J. Endobronchial valves for emphysema without interlobar collateral ventilation. *N Engl J Med*. 2015;373(24):2325–35. <https://doi.org/10.1056/NEJMoa1507807>
- 28 Broadbush VC, Ernst JD, Jr TEK, Lazarus SC, Sarmiento KF, Schnapp LM, et al. Chapter 44. Hypoxemia. Murray & Nadel's Textbook of Respiratory Medicine E-Book. Elsevier Health Sciences; 2021.
- 29 Long term domiciliary oxygen therapy in chronic hypoxic cor pulmonale complicating chronic bronchitis and emphysema. Report of the medical research council working party. *Lancet*. 1981;1(8222):681–6.
- 30 NOCTURNAL OXYGEN THERAPY TRIAL GROUP*. Continuous or nocturnal oxygen therapy in hypoxemic chronic obstructive lung disease: a clinical trial. Nocturnal oxygen therapy trial group. *Ann Intern Med*. 1980;93(3):391–8. <https://doi.org/10.7326/0003-4819-93-3-391>
- 31 Sciruba FC, Ernst A, Herth FJF, Strange C, Criner GJ, Marquette CH, et al. A randomized study of endobronchial valves for advanced emphysema. *N Engl J Med*. 2010;363(13):1233–44. <https://doi.org/10.1056/NEJMoa0900928>
- 32 Davey C, Zoumot Z, Jordan S, Carr DH, Polkey MI, Shah PL, et al. Bronchoscopic lung volume reduction with endobronchial valves for patients with heterogeneous emphysema and intact interlobar fissures (the BeLieVeR-HiFi trial): study design and rationale. *Thorax*. 2015;70(3):288–90. <https://doi.org/10.1136/thoraxjnl-2014-205127>
- 33 Li S, Wang G, Wang C, Gao X, Jin F, Yang H, et al. The REACH trial: a randomized controlled trial assessing the safety and effectiveness of the spiration® valve system in the treatment of severe emphysema. *Respiration*. 2019;97(5):416–27. <https://doi.org/10.1159/000494327>
- 34 Lenga P, Grah C, Ruwwe-Glösenkamp C, Saccomanno J, Rückert J, Eggeling S, et al. Endoscopic lung volume reduction with one-way valves in patients with severe chronic obstructive pulmonary disease with hypercapnia. *Respiration*. 2022;101(9):823–32. <https://doi.org/10.1159/000524996>
- 35 Roetting M, Kriegsmann K, Polke M, Polke N, Kontogianni K, Eberhardt R, et al. Endoscopic valve therapy in COPD patients with hypercapnia. *Respiration*. 2022;101(10):918–24. <https://doi.org/10.1159/000525729>
- 36 Cobb K, Kenyon J, Lu J, Krieger B, Perelas A, Nana-Sinkam P, et al. COPD is associated with increased cardiovascular disease risk independent of phenotype. *Respirology*. 2024;29(12):1047–57. <https://doi.org/10.1111/resp.14799>
- 37 Xu S, Gu Z, Zhu W, Feng S. Association of COPD with adverse outcomes in heart failure patients with preserved ejection fraction. *ESC Heart Fail*. 2025;12(2):799–808. <https://doi.org/10.1002/ehf2.14958>
- 38 Ather S, Chan W, Bozkurt B, Aguilar D, Ramasubbu K, Zachariah AA, et al. Impact of noncardiac comorbidities on morbidity and mortality in a predominantly male population with heart failure and preserved versus reduced ejection fraction. *J Am Coll Cardiol*. 2012;59(11):998–1005. <https://doi.org/10.1016/j.jacc.2011.11.040>
- 39 Writing Committee Members, Thompson A, Fleischmann KE, Smailowitz NR, de Las Fuentes L, Mukherjee D, et al. 2024 AHA/ACC/ACS/ASNC/HRS/SCA/SCCT/SCMR/SVM Guideline for perioperative cardiovascular management for noncardiac surgery: a report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines. *J Am Coll Cardiol*. 2024;84(19):1869–969. <https://doi.org/10.1016/j.jacc.2024.06.013>
- 40 van der Molen MC, Hartman JE, Vanfleteren LEGW, Kerstjens HAM, van Melle JP, Willems TP, et al. Reduction of lung hyperinflation improves cardiac preload, contractility, and output in Emphysema: a clinical trial in patients who received endobronchial valves. *Am J Respir Crit Care Med*. 2022;206(6):704–11. <https://doi.org/10.1164/rccm.202201-0214OC>
- 41 Pizarro C, Schueler R, Hammerstingl C, Tuleta I, Nickenig G, Skowasch D. Impact of endoscopic lung volume reduction on right ventricular myocardial function. *PLoS One*. 2015;10(4):e0121377. <https://doi.org/10.1371/journal.pone.0121377>
- 42 Andersen KH, Iversen M, Kjaergaard J, Mortensen J, Nielsen-Kudsk JE, Bendstrup E, et al. Prevalence, predictors, and survival in pulmonary hypertension related to end-stage chronic obstructive pulmonary disease. *J Heart Lung Transpl*. 2012;31(4):373–80. <https://doi.org/10.1016/j.healun.2011.11.020>

- 43 Ter Haar EAMD, Slebos D-J, Klooster K, Pouwels SD, Hartman JE. Comorbidities reduce survival and quality of life in COPD with severe lung hyperinflation. *ERJ Open Res.* 2024;10(6):00268-2024-2024. <https://doi.org/10.1183/23120541.00268-2024>
- 44 Zhang L, Liu Y, Zhao S, Wang Z, Zhang M, Zhang S, et al. The incidence and prevalence of pulmonary hypertension in the COPD population: a systematic review and meta-analysis. *Int J Chron Obstruct Pulmon Dis.* 2022;17:1365-79. <https://doi.org/10.2147/COPD.S359873>
- 45 Shlobin OA, Adir Y, Barbera JA, Cottin V, Harari S, Jutant EM, et al. Pulmonary hypertension associated with lung diseases. *Eur Respir J.* 2024;64(4):2401200. <https://doi.org/10.1183/13993003.01200-2024>
- 46 Humbert M, Kovacs G, Hoeper MM, Badagliacca R, Berger RMF, Brida M, et al. 2022 ESC/ERS Guidelines for the diagnosis and treatment of pulmonary hypertension. *Eur Respir J.* 2023;61(1):2200879. <https://doi.org/10.1183/13993003.00879-2022>
- 47 Valipour A, Slebos D-J, Herth F, Darwiche K, Wagner M, Ficker JH, et al. Endobronchial valve therapy in patients with homogeneous emphysema. Results from the IMPACT Study. *Am J Respir Crit Care Med.* 2016;194(9):1073-82. <https://doi.org/10.1164/rccm.201607-1383OC>
- 48 Kemp SV, Slebos D-J, Kirk A, Kornaszewska M, Carron K, Ek L, et al. A multicenter randomized controlled trial of Zephyr endobronchial valve treatment in heterogeneous Emphysema (TRANSFORM). *Am J Respir Crit Care Med.* 2017;196(12):1535-43. <https://doi.org/10.1164/rccm.201707-1327OC>
- 49 Eberhardt R, Gerovasili V, Kontogianni K, Gompelmann D, Ehlken N, Herth FJF, et al. Endoscopic lung volume reduction with endobronchial valves in patients with severe emphysema and established pulmonary hypertension. *Respiration.* 2015;89(1):41-8. <https://doi.org/10.1159/000368369>
- 50 Fiorelli A, Cascone R, Natale G, Peritore V, Vanni C, Poggi C, et al. Cardio-Pulmonary changes after bronchoscopic lung volume reduction with endobronchial one-way valves. *Lung.* 2020;198(3):565-73. <https://doi.org/10.1007/s00408-020-00351-4>
- 51 Weg IL, Rossoff L, McKeon K, Michael Graver L, Scharf SM. Development of pulmonary hypertension after lung volume reduction surgery. *Am J Respir Crit Care Med.* 1999;159(2):552-6. <https://doi.org/10.1164/ajrccm.159.2.9802056>
- 52 Criner GJ, Scharf SM, Falk JA, Gaughan JP, Sternberg AL, Patel NB, et al. Effect of lung volume reduction surgery on resting pulmonary hemodynamics in severe emphysema. *Am J Respir Crit Care Med.* 2007;176(3):253-60. <https://doi.org/10.1164/rccm.200608-1114OC>
- 53 Thuppal S, Crabtree T, Markwell S, Colle J, Boley T, Manning B, et al. Pulmonary hypertension: a contraindication for lung volume reduction surgery? *Ann Thorac Surg.* 2020;109(3):902-6. <https://doi.org/10.1016/j.athoracsur.2019.09.023>
- 54 Haniuda M, Kubo K, Fujimoto K, Aoki T, Yamada T, Amano J. Different effects of lung volume reduction surgery and lobectomy on pulmonary circulation. *Ann Surg.* 2000;231(1):119-25. <https://doi.org/10.1097/0000658-200001000-00017>
- 55 Caviezel C, Aruldas C, Franzen D, Ulrich S, Inci I, Schneider D, et al. Lung volume reduction surgery in selected patients with emphysema and pulmonary hypertension. *Eur J Cardio Thorac Surg.* 2018;54(3):565-71. <https://doi.org/10.1093/ejcts/ezy092>
- 56 Mahajan AK, Collar N, Muldowney F, Gorka P, Duong DK, Patel PP, et al. Echocardiography findings for pulmonary hypertension during workup for bronchoscopic lung volume reduction. *J Bronchology Interv Pulmonol.* 2025;32(1):e0997. <https://doi.org/10.1097/LBR.0000000000000997>
- 57 Celli BR, Cote CG, Marin JM, Casanova C, Montes de Oca M, Mendez RA, et al. The body-mass index, airflow obstruction, dyspnea, and exercise capacity index in chronic obstructive pulmonary disease. *N Engl J Med.* 2004;350(10):1005-12. <https://doi.org/10.1056/NEJMoa021322>
- 58 Benz E, Trajanoska K, Lahousse L, Schoufour JD, Terzikhan N, De Roos E, et al. Sarcopenia in COPD: a systematic review and meta-analysis. *Eur Respir Rev.* 2019;28(154):190049. <https://doi.org/10.1183/16000617.0049-2019>
- 59 Ter Haar EAMD, Slebos D-J, Hartman JE. Extrapulmonary effects of lung volume reduction in severe emphysema: a systematic review. *Eur Respir Rev.* 2025;34(176):240258. <https://doi.org/10.1183/16000617.0258-2024>
- 60 Fernandez-Bussy S, Kornafeld A, Labarca G, Abia-Trujillo D, Patel NM, Herth FJF. Endoscopic lung volume reduction in relation to body mass index in patients with severe heterogeneous emphysema. *Respiration.* 2020;99(6):477-83. <https://doi.org/10.1159/000507591>
- 61 Vaughan P, Oey IF, Steiner MC, Morgan MDL, Waller DA. A prospective analysis of the inter-relationship between lung volume reduction surgery and body mass index. *Eur J Cardio Thorac Surg.* 2007;32(6):839-42. <https://doi.org/10.1016/j.ejcts.2007.09.016>
- 62 Takahashi M, Fukuoka J, Nitta N, Takazakura R, Nagatani Y, Murakami Y, et al. Imaging of pulmonary emphysema: a pictorial review. *Int J Chron Obstruct Pulmon Dis.* 2008;3(2):193-204. <https://doi.org/10.2147/copd.s2639>
- 63 Smith BM, Austin JHM, Newell JD, D'Souza BM, Rozenshtein A, Hoffman EA, et al. Pulmonary emphysema subtypes on computed tomography: the MESA COPD study. *Am J Med.* 2014;127(1):94.e7-23. <https://doi.org/10.1016/j.amjmed.2013.09.020>
- 64 Stoller JK, Aboussouan LS. A review of a 1-antitrypsin deficiency. *Am J Respir Crit Care Med.* 2012;185(3):246-59. <https://doi.org/10.1164/rccm.201108-1428CI>
- 65 Camp PC, Sugarbaker DJ. Surgical interventions for emphysema. *Semin Thorac Cardiovasc Surg.* 2007;19(2):157-71. <https://doi.org/10.1053/j.semtcvs.2007.05.004>
- 66 Marchetti N, Criner GJ. Surgical approaches to treating emphysema: Lung volume reduction surgery, bullectomy, and lung transplantation. *Semin Respir Crit Care Med.* 2015;36(4):592-608. <https://doi.org/10.1055/s-0035-1556064>
- 67 Lee EG, Rhee CK. Bronchoscopic lung volume reduction using an endobronchial valve to treat a huge emphysematous bullae: a case report. *BMC Pulm Med.* 2019;19(1):92. <https://doi.org/10.1186/s12890-019-0849-z>
- 68 Tian Q, An Y, Xiao B-B, Chen L-A. Treatment of giant emphysematous bulla with endobronchial valves in patients with chronic obstructive pulmonary disease: a case series. *J Thorac Dis.* 2014;6(12):1674-80. <https://doi.org/10.3978/j.issn.2072-1439.2014.11.07>
- 69 Koster TD, Shah PL, Valipour A, Criner GJ, Herth FJF, Sue R, et al. Optimizing clinical outcomes for bronchoscopic lung volume reduction with zephyr® valves. *Respir Med.* 2024;227:107639. <https://doi.org/10.1016/j.rmed.2024.107639>
- 70 van Geffen WH, Klooster K, Hartman JE, Ten Hacken NHT, Kerstjens HAM, Wolf RFE, et al. Pleural adhesion assessment as a predictor for pneumothorax after endobronchial valve treatment. *Respiration.* 2017;94(2):224-31. <https://doi.org/10.1159/000477258>
- 71 Gompelmann D, Benjamin N, Kontogianni K, Herth F, Heussel CP, Hoffmann H, et al. Clinical and radiological outcome following pneumothorax after endoscopic lung volume reduction with valves. *Int J Chron Obstruct Pulmon Dis.* 2016;11:3093-9. <https://doi.org/10.2147/COPD.S117890>
- 72 Everaerts S, Hartman JE, Van Dijk M, Koster TD, Slebos D-J, Klooster K. Bronchoscopic lung volume reduction in patients with emphysema due to Alpha-1 antitrypsin deficiency. *Respiration.* 2023;102(2):134-42. <https://doi.org/10.1159/000528182>
- 73 Hillerdal G, Mindus S. One-to four-year follow-up of endobronchial lung volume reduction in alpha-1-antitrypsin deficiency patients: a case series. *Respiration.* 2014;88(4):320-8. <https://doi.org/10.1159/000365662>
- 74 Hogarth DK, Delage A, Zgoda MA, Nsiahdosu S, Himes D, Reed MF. Efficacy and safety of the spiration valve System™ for the treatment of severe emphysema in patients with Alpha-1 antitrypsin deficiency (EMPROVE). *Respir Med.* 2024;224:107565. <https://doi.org/10.1016/j.rmed.2024.107565>

- 75 Hartman JE, Criner GJ, Moore WH, van Rikxoort EM, Sciurba FC, Shah PL, et al. HRCT characteristics of severe emphysema patients: interobserver variability among expert readers and comparison with quantitative software. *Eur J Radiol.* 2021; 136:109561. <https://doi.org/10.1016/j.ejrad.2021.109561>
- 76 Gevenois PA, De Vuyst P, Sy M, Scillia P, Chaminade L, de Maertelaer V, et al. Pulmonary emphysema: quantitative CT during expiration. *Radiology.* 1996;199(3): 825–9. <https://doi.org/10.1148/radiology.199.3.8638012>
- 77 Crossley D, Renton M, Khan M, Low EV, Turner AM. CT densitometry in emphysema: a systematic review of its clinical utility. *Int J Chron Obstruct Pulmon Dis.* 2018;13:547–63. <https://doi.org/10.2147/COPD.S143066>
- 78 Hartman JE, Roodenburg SA, van Dijk M, Koster TD, Klooster K, Slebos D-J. Response to endobronchial valve treatment: it's all about the target lobe. *ERJ Open Res.* 2023;9(4):00155–2023–2023. <https://doi.org/10.1183/23120541.00155-2023>
- 79 Herth FJF, Kirby M, Sieren J, Herth J, Schirm J, Wood S, et al. The modern art of reading computed tomography images of the lungs: quantitative CT. *Respiration.* 2018;95(1):8–17. <https://doi.org/10.1159/000480435>
- 80 Ramaswamy A, Puchalski J. Bronchoscopic lung volume reduction: recent updates. *J Thorac Dis.* 2018;10(4):2519–27. <https://doi.org/10.21037/jtd.2018.02.72>
- 81 Klooster K, Koster TD, Ruwwe-Glücksenkamp C, Theilig D, Doellinger F, Saccomanno J, et al. An integrative approach of the fissure completeness score and chartis assessment in endobronchial valve treatment for emphysema. *Int J Chron Obstruct Pulmon Dis.* 2020;15:1325–34. <https://doi.org/10.2147/COPD.S242210>
- 82 de Oliveira HG, de Oliveira SM, Rambo RR, de Macedo Neto AV. Fissure integrity and volume reduction in emphysema: a retrospective study. *Respiration.* 2016; 91(6):471–9. <https://doi.org/10.1159/000446288>
- 83 Argula RG, Strange C, Ramakrishnan V, Goldin J. Baseline regional perfusion impacts exercise response to endobronchial valve therapy in advanced pulmonary emphysema. *Chest.* 2013;144(5):1578–86. <https://doi.org/10.1378/chest.12-2826>
- 84 Thomsen C, Theilig D, Herzog D, Poellinger A, Doellinger F, Schreiter N, et al. Lung perfusion and emphysema distribution affect the outcome of endobronchial valve therapy. *Int J Chron Obstruct Pulmon Dis.* 2016;11:1245–59. <https://doi.org/10.2147/COPD.S101003>
- 85 Ide Bolet S, Sisti J, Cheng K, Dadparvar S. Quantitative lobar Tc99m-MAA SPECT/CT of the lung in pre- and post-procedural guidance for bronchoscopic lung volume reduction. *Nucl Med Commun.* 2024;45(9): 770–8. <https://doi.org/10.1097/MNM.0000000000001868>
- 86 Kristiansen JF, Perch M, Iversen M, Krakauer M, Mortensen J. Lobar quantification by ventilation/perfusion SPECT/CT in patients with severe emphysema undergoing lung volume reduction with endobronchial valves. *Respiration.* 2019;98(3):230–8. <https://doi.org/10.1159/000500407>
- 87 Gietema HA, Walraven KHM, Posthuma R, Mitea C, Slebos D-J, Vanfleteren LEGW. Dual-Energy computed tomography compared to lung perfusion scintigraphy to assess pulmonary perfusion in patients screened for endoscopic lung volume reduction. *Respiration.* 2021;100(12):1186–95. <https://doi.org/10.1159/000517598>
- 88 Tee VST, Nguyen P, Jersmann H, Grosser D, Crouch B, Lorraine B, et al. Use of ventilation-perfusion single-photon emission computed tomography to select the target lobe for endobronchial valve lung volume reduction. *Respiration.* 2021; 100(9):886–97. <https://doi.org/10.1159/000515336>
- 89 Borgheresi A, Cesari E, Agostini A, Badaloni M, Balducci S, Tola E, et al. Pulmonary emphysema: the assessment of lung perfusion with dual-energy CT and pulmonary scintigraphy. *Radiol Med.* 2024;129(11): 1622–32. <https://doi.org/10.1007/s11547-024-01883-y>
- 90 Roodenburg SA, Klooster K, Slebos D-J, Hartman JE. The impact of emphysema heterogeneity on treatment response after endobronchial valve treatment. *ERJ Open Res.* 2023;9(4):00279–2023–2023. <https://doi.org/10.1183/23120541.00279-2023>
- 91 Hetzel J, Boeckeler M, Horger M, Ehab A, Kloth C, Wagner R, et al. A new functional method to choose the target lobe for lung volume reduction in emphysema - comparison with the conventional densitometric method. *Int J Chron Obstruct Pulmon Dis.* 2017;12:2621–8. <https://doi.org/10.2147/COPD.S139304>
- 92 Wang Y, Lai T-W, Xu F, Zhou J-S, Li Z-Y, Xu X-C, et al. Efficacy and safety of bronchoscopic lung volume reduction therapy in patients with severe emphysema: a meta-analysis of randomized controlled trials. *Oncotarget.* 2017;8(44): 78031–43. <https://doi.org/10.18632/oncotarget.19352>
- 93 Costa Filho FF, Buckley JD, Furlan A, Campbell S, Hickok K, Kroth PJ. Inpatient complication rates of bronchoscopic lung volume reduction in the United States. *Chest.* 2025;167(2):436–43. <https://doi.org/10.1016/j.chest.2024.08.012>
- 94 Brock JM, Dittrich SA, Eichhorn F, Schlamp K, Kontogianni K, Herth FJF. Lessons learned: risk factors and clinical impact of severe pneumothorax after endoscopic lung volume reduction with endobronchial valves. *Chest.* 2024;S0012-3692(24):05456–4.
- 95 Gompelmann D, Lim H-J, Eberhardt R, Gerovasili V, Herth FJ, Heussel CP, et al. Predictors of pneumothorax following endoscopic valve therapy in patients with severe emphysema. *Int J Chron Obstruct Pulmon Dis.* 2016;11:1767–73. <https://doi.org/10.2147/COPD.S106439>
- 96 van Dijk M, Sue R, Criner GJ, Gompelmann D, Herth FJF, Hogarth DK, et al. Expert statement: pneumothorax associated with one-way valve therapy for emphysema: 2020 update. *Respiration.* 2021;100(10): 969–78. <https://doi.org/10.1159/000516326>
- 97 Valipour A, Slebos D-J, Oliveira HG, Eberhardt R, Freitag L, Criner GJ, et al. Expert statement: pneumothorax associated with endoscopic valve therapy for emphysema - potential mechanisms, treatment algorithm, and case examples. *RES.* 2014; 87(6):513–21.
- 98 Mahajan AK, Collar N, Muldowney F, Ochoa Cuba MA, Wang H, Shah N, et al. Incidence and outcomes for post-bronchoscopic lung volume reduction (BLVR) pneumothoraces stratified by target lobe treated. *Respir Med.* 2025;243:108148. <https://doi.org/10.1016/j.rmed.2025.108148>
- 99 Herth FJF, Eberhardt R, Gompelmann D, Ficker JH, Wagner M, Ek L, et al. Radiological and clinical outcomes of using Chartis™ to plan endobronchial valve treatment. *Eur Respir J.* 2013;41(2):302–8. <https://doi.org/10.1183/09031936.00015312>
- 100 Gompelmann D, Eberhardt R, Slebos D-J, Brown MS, Abtin F, Kim HJ, et al. Diagnostic performance comparison of the chartis system and high-resolution computerized tomography fissure analysis for planning endoscopic lung volume reduction. *Respirology.* 2014;19(4): 524–30. <https://doi.org/10.1111/resp.12253>
- 101 Zephyr endobronchial valve system: instructions for use. [Internet]. [cited 2025 Mar 18]. Available from: https://www.accessdata.fda.gov/cdrh_docs/pdf18/P180002C.pdf
- 102 Roodenburg SA, Klooster K, Hartman JE, Koster TD, van Dijk M, Slebos D-J. Revision bronchoscopy after endobronchial valve treatment for emphysema: indications, findings and outcomes. *Int J Chron Obstruct Pulmon Dis.* 2021;16: 1127–36. <https://doi.org/10.2147/COPD.S302662>
- 103 Mahajan AK, Collar N, Muldowney F, Patel PP, Hogarth DK, Duong DK. Incidence and outcomes of revision bronchoscopies following Bronchoscopic Lung Volume Reduction (BLVR). *J Bronchology Interv Pulmonol.* 2025; 32(1):e1002. <https://doi.org/10.1097/LBR.0000000000001002>

- 104 Brock JM, Schuster PU, Böhmker F, Eberhardt R, Gompelmann D, Kontogianni K, et al. Endobronchial valve replacements in patients with advanced emphysema after endoscopic lung volume reduction. *Int J Chron Obstruct Pulmon Dis.* 2023;18:933–43. <https://doi.org/10.2147/COPD.S408674>
- 105 Welling JBA, Hartman JE, van Rikxoort EM, Ten Hacken NHT, Kerstjens HAM, Klooster K, et al. Minimal important difference of target lobar volume reduction after endobronchial valve treatment for emphysema. *Respirology.* 2018;23(3):306–10. <https://doi.org/10.1111/resp.13178>
- 106 Wienker J, Darwiche K, Wälscher J, Winantea J, Hagemann M, Büscher E, et al. Clinical impact of compensatory hyperinflation of the nontreated adjacent lobe after bronchoscopic lung volume reduction with valves. *Int J Chron Obstruct Pulmon Dis.* 2022;17:1523–36. <https://doi.org/10.2147/COPD.S364448>
- 107 Koster TD, Klooster K, Ten Hacken NHT, van Dijk M, Slebos D-J. Endobronchial valve therapy for severe emphysema: an overview of valve-related complications and its management. *Expert Rev Respir Med.* 2020;14(12):1235–47. <https://doi.org/10.1080/17476348.2020.1813571>
- 108 Brown MS, Kim HJ, Abtin FG, Strange C, Galperin-Aizenberg M, Pais R, et al. Emphysema lung lobe volume reduction: effects on the ipsilateral and contralateral lobes. *Eur Radiol.* 2012;22(7):1547–55. <https://doi.org/10.1007/s00330-012-2393-6>
- 109 Winantea J, Stiehl K, Karpf-Wissel R, Funke F, Hautzel H, Schwarz B, et al. Impact of bronchoscopic lung volume reduction with valves on the pulmonary gas exchange. *J Clin Med.* 2024;13(8):2354. <https://doi.org/10.3390/jcm13082354>
- 110 Alsheimer KM, Burgei JW, Lantry JF. Physiologic changes resulting in low-flow states mimicking a pulmonary embolism following endobronchial valve placement for lung volume reduction. *Chest.* 2024;166(4):A5287–8. <https://doi.org/10.1016/j.chest.2024.06.3146>
- 111 van der Molen MC, Posthuma R, Hartman JE, van der Vaart H, Bij de Vaate E, Vaes AW, et al. Impact and timing of pulmonary rehabilitation in patients undergoing bronchoscopic lung volume reduction with endobronchial valves: a multicentre randomized controlled trial in patients with severe emphysema. *Respirology.* 2024;29(8):694–703. <https://doi.org/10.1111/resp.14734>
- 112 Brock JM, Kontogianni K, Sciruba FC, Criner GJ, Herth F. Utility of rehabilitation prior to bronchoscopic lung volume reduction: post hoc analysis of the VENT trial. *ERJ Open Res.* 2024;10(1):00735–2023–2023. <https://doi.org/10.1183/23120541.00735-2023>
- 113 McCarthy B, Casey D, Devane D, Murphy K, Murphy E, Lacasse Y. Pulmonary rehabilitation for chronic obstructive pulmonary disease. *Cochrane Database Syst Rev.* 2015;2015(2):CD003793. <https://doi.org/10.1002/14651858.CD003793.pub3>
- 114 2025 GOLD Report [Internet]. Global Initiative for Chronic Obstructive lung disease - GOLD. [cited 2025 Mar 4]. Available from: <https://goldcopd.org/2025-gold-report/>
- 115 Fiorelli A, D'Andrilli A, Anile M, Diso D, Poggi C, Polverino M, et al. Sequential bilateral bronchoscopic lung volume reduction with one-way valves for heterogeneous emphysema. *Ann Thorac Surg.* 2016;102(1):287–94. <https://doi.org/10.1016/j.athoracsur.2016.02.018>
- 116 Guibert N, Fumat R, Dupuis M, Dusselier M, Favard F, Heluain V, et al. Bilateral endobronchial valves treatment for severe emphysema. *ERJ Open Res.* 2022;8(2):00672–2021–2021. <https://doi.org/10.1183/23120541.00672-2021>
- 117 Trudzinski FC, Lepper PM, Leppert D, Langer F, Lensch C, Flaig M, et al. Bilateral endoscopic lung volume reduction in patients with severe emphysema. *Respiration.* 2016;92(5):356–8. <https://doi.org/10.1159/000450758>
- 118 Ninane V, Geltner C, Bezzi M, Foccoli P, Gottlieb J, Welte T, et al. Multicentre european study for the treatment of advanced emphysema with bronchial valves. *Eur Respir J.* 2012;39(6):1319–25. <https://doi.org/10.1183/09031936.00019711>
- 119 Egenod T, Guibert N, Ammar Y, Kessler R, Toubanc B, Favrolt N, et al. Endobronchial valves: 1st multicenter retrospective study on the 2-step approach. *Respir Med Res.* 2023;83:100957. <https://doi.org/10.1016/j.resmer.2022.100957>
- 120 Egenod T, Tricard J, Fumat R, Simonneau Y, Favard F, Guillot M-S, et al. Two-Stage bronchoscopic endobronchial valve treatment can lead to progressive lung volume reduction and may decrease pneumothorax risk. *Int J Chron Obstruct Pulmon Dis.* 2021;16:1957–65. <https://doi.org/10.2147/COPD.S307829>
- 121 Lentz RJ, Low S-W, Saetlele T, Rickman OB, Aboudara M, Maldonado F. Association between inspired oxygen fraction and pneumothorax after endobronchial valve placement for emphysema. *Ann ATS.* 2023;20(6):926–9. <https://doi.org/10.1513/AnnalsATS.202208-664RL>